Wireless Weather Station Technical Manual

Release 1.0

Document Control

Item	Value
Document Title	Wireless Weather Station Technical Manual
Document ID	WWS-TM-001
Revision	1.0
Status	Release
Author	Jim
Classification	Technical Reference
System Version	Release 1 Platform

Revision History

Revision	Date	Description
0.1	Initial Development	Original weather station concept
0.5	Development Phase	Multi-controller architecture established
0.6	Development Phase	LoRa communications implemented
0.7	Development Phase	Solar-powered outdoor node added
0.8	Development Phase	Remote reboot and uptime monitoring added
0.9	Development Phase	Dashboard and database architecture expanded
1.0	October 2026	Initial technical manual release

Acronyms and Definitions

Acronym	Definition
ADC	Analog-to-Digital Converter
BMP280	Bosch Barometric Pressure Sensor
DS3231	Real-Time Clock Module
EEPROM	Electrically Erasable Programmable Read-Only Memory
ESP32	Espressif 32-bit Microcontroller
GPIO	General Purpose Input / Output
HTTP	Hypertext Transfer Protocol
I^2C	Inter-Integrated Circuit
LCD	Liquid Crystal Display
LoRa	Long Range Radio
RTC	Real-Time Clock
SHT31-D	Temperature/Humidity Sensor
UART	Universal Asynchronous Receiver / Transmitter
WU	Weather Underground

List of Figures

Figure	Description
3-1	System Architecture Diagram
4-1	I^2C Sensor Bus Diagram
4-2	Sensors inside Radiation Shield

4-3	Underside of Radiation Shield Perfboard
4-4	Radiation Shield Installation
4-5	Indoor Electronics Diagram
4-6	Indoor Controller Assembly
4-7	Outdoor Electronics Diagram
4-8	Outdoor Sensor Mast
4-9	Solar Power System
4-10	Outdoor Solar Power Assembly
4-11	Outdoor Controller Electronics Assembly
6-1	Dashboard Data Flow
8-1	Communications and Data Acquisition Architecture
10-1	Dashboard Screenshot
11-1	Environmental Processing Pipeline
11-2	Temperature Suppression Curve
11-3	Dew Point Calculation Flow
12-1	Outdoor Sensor Mast Installation
12-2	Radiation Shield Installation
13-1	Troubleshooting Workflow
14-1	Annual Maintenance Verification Flow
15-1	Cold Start Initialization Sequence

List of Tables

Table	Description
3-1	Controller Responsibility Matrix
4-1	Indoor Hardware Inventory
4-2	Outdoor Hardware Inventory
4-3	Arduino Mega GPIO Assignments
4-4	ESP32-S3 GPIO Assignments
4-5	ESP32-C3 GPIO Assignments
4-6	Sensor Wiring Summary
4-7	Solar Power Components

Section 1 – Introduction

1.1 Purpose

This manual documents the design, architecture, operation, maintenance, and future development of the Wireless Weather Station system.

The system combines environmental sensing, local monitoring, wireless telemetry, Internet connectivity, database storage, and dashboard presentation into a fully integrated weather-monitoring platform.

The document is intended to serve as the primary engineering reference for:

- Construction
- Maintenance
- Troubleshooting
- Firmware modification
- Future expansion

1.2 System Description

The weather station is built around a distributed architecture based on three cooperating controllers:

Arduino Mega

The Arduino Mega functions as the primary supervisory controller.

Responsibilities include:

- User interface management
- Sensor integration
- EEPROM management
- RTC management
- Min/max tracking
- Outdoor-node supervision
- Data aggregation

ESP32-S3 Gateway

The ESP32-S3 serves as the Internet gateway and presentation platform.

Responsibilities include:

- Dashboard hosting
- Wi-Fi communications
- Database uploads
- Weather Underground uploads
- Status monitoring

ESP32-C3 Outdoor Node

The ESP32-C3 serves as the outdoor telemetry controller.

Responsibilities include:

- Wind-speed measurement
- Wind-direction processing
- Rainfall measurement

- LoRa communications
- Outdoor uptime tracking

1.3 Design Goals

The weather station was designed around several key objectives:

Reliability

The station must continue operating through temporary communications failures and Internet outages.

Serviceability

The system should be diagnosable without specialized equipment.

Modularity

Major subsystems must be replaceable independently.

Expandability

Future sensors and capabilities should be incorporated without redesigning the architecture.

Data Integrity

Weather observations and system status information must be preserved for analysis and long-term review.

1.4 Major Features

The current system provides:

- Temperature monitoring
- Relative humidity monitoring
- Dew-point calculation
- Barometric pressure measurement
- Wind-speed measurement
- Wind-gust measurement
- Wind-direction measurement

- Rainfall measurement
- Historical data storage
- Browser-based dashboard
- Weather Underground integration
- Automatic fault recovery
- Solar-powered outdoor operation

Section 2 – System Overview

2.1 System Functions

The station performs four major functions:

Measure

Transmit

Display

Archive

Environmental data is acquired from indoor and outdoor sensors, transmitted through internal communications links, displayed locally and remotely, and archived within a database.

2.2 Environmental Measurements

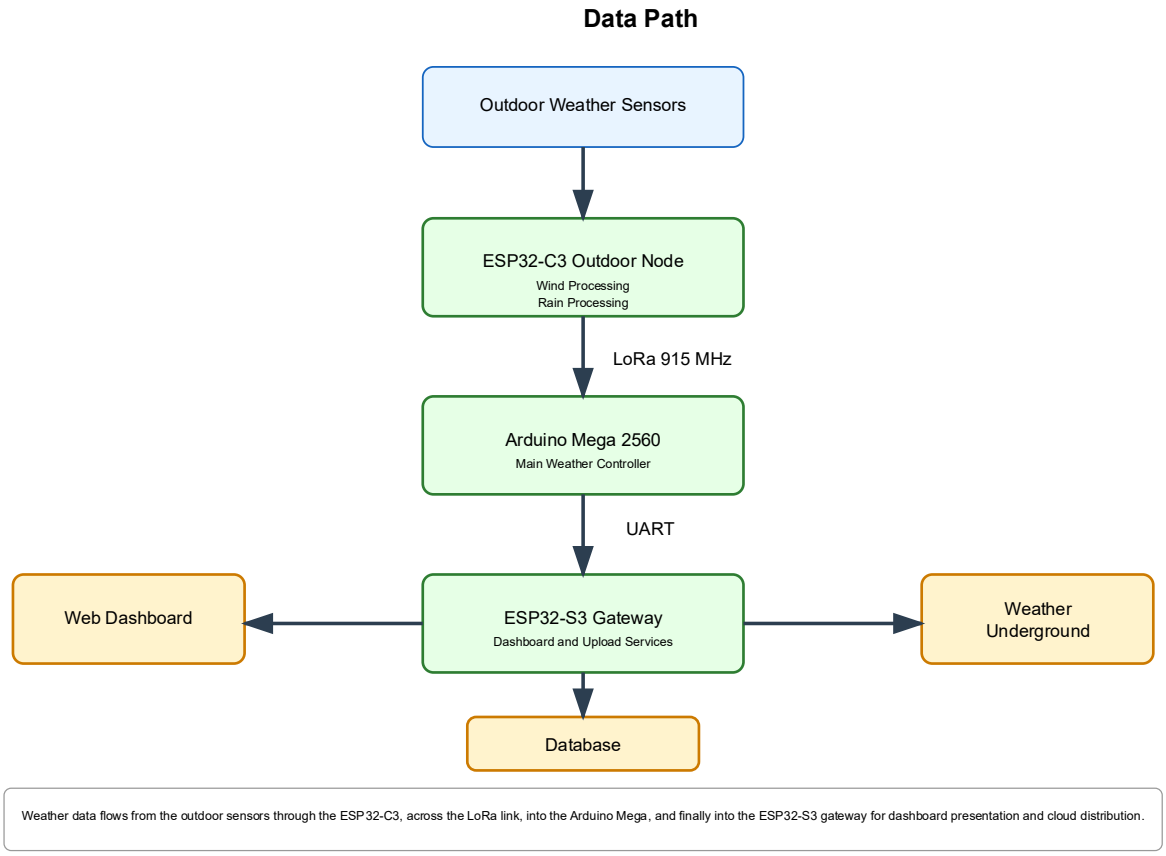
The station currently measures:

Parameter	**Source**
Temperature	SHT31-D / BMP280
Humidity	SHT31-D
Dew Point	Calculated
Pressure	BMP280
Wind Speed	Anemometer

Wind Gust	Anemometer
Wind Direction	Wind Vane
Rainfall Rate	Rain Gauge
Daily Rainfall	Rain Gauge
Outdoor Uptime	ESP32-C3

2.3 Data Path

Environmental measurements traverse the following path:



This architecture separates measurement from presentation and storage.

2.4 Operational Cycle

The normal operating cycle is:

1. Outdoor node acquires measurements.
2. Outdoor node transmits LoRa packet.
3. Arduino Mega parses telemetry.
4. Arduino Mega combines indoor and outdoor data.
5. Mega generates weather packet.

6. ESP32-S3 receives packet.
7. Dashboard updated.
8. Database updated.
9. Weather Underground updated.

The cycle typically repeats every five seconds.

Section 3 – System Architecture

3.1 Architectural Overview

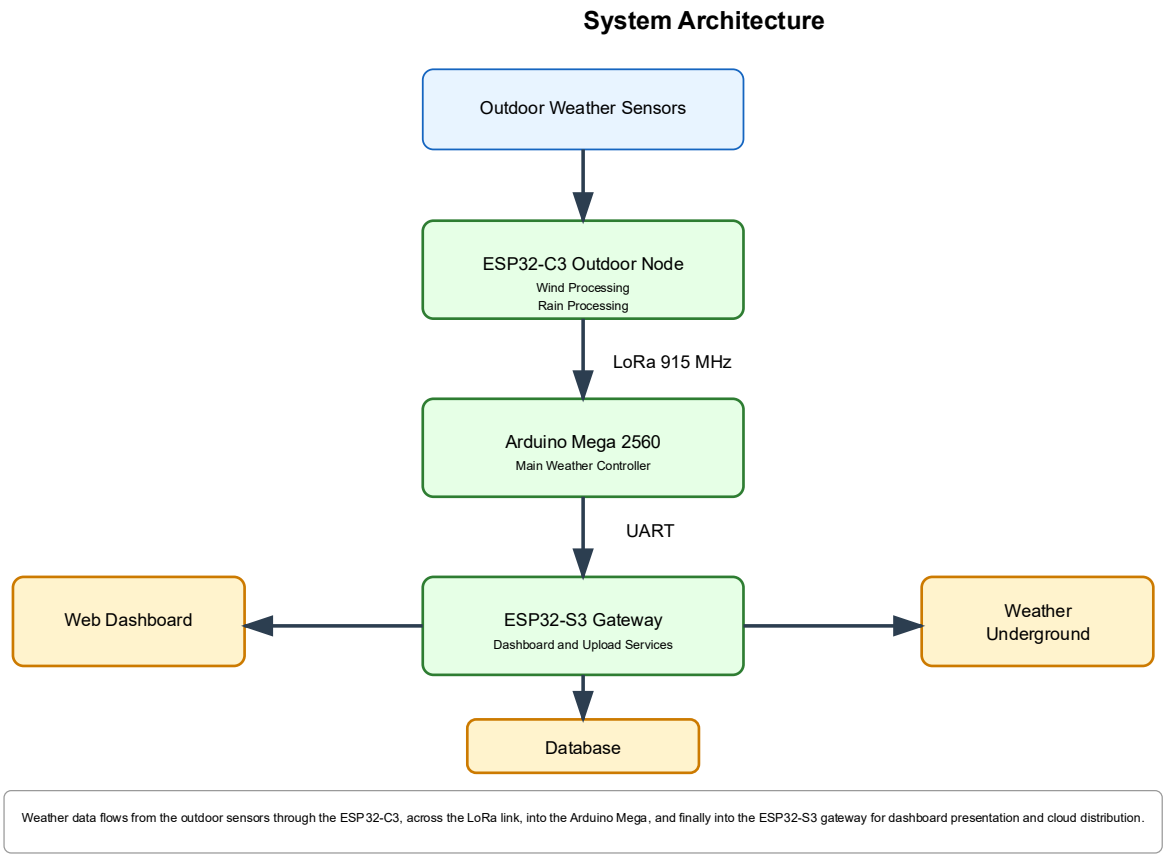
The weather station utilizes a three-controller distributed architecture.

This design distributes responsibilities across specialized controllers rather than placing all functions on a single device.

Advantages include:

- Improved reliability
- Lower software complexity
- Easier debugging
- Improved scalability

Figure 3-1. System Architecture Diagram



3.2 Controller Responsibility Matrix

Controller	Primary Responsibilities
Arduino Mega	User interface, sensing, recovery management
ESP32-S3	Dashboard, Wi-Fi, uploads
ESP32-C3	Outdoor measurements and telemetry

3.3 Indoor Controller Architecture

The indoor controller subsystem consists of:

- Arduino Mega
- LCD Keypad Shield
- SHT31-D
- BMP280
- DS3231 RTC
- LoRa Module
- ESP32-S3

The Mega acts as the central authority for environmental processing and operational management.

3.4 Outdoor Controller Architecture

The outdoor subsystem contains:

- ESP32-C3
- LoRa module
- Wind vane
- Anemometer
- Rain gauge
- Solar power subsystem

This subsystem operates autonomously and transmits weather observations to the indoor station.

3.5 Fault Tolerance

The architecture contains multiple layers of fault tolerance:

Sensor Redundancy

Temperature can be obtained from multiple sources.

Communications Monitoring

LoRa communication health is continuously monitored.

Automatic Recovery

The Mega can command remote reboots of the outdoor node.

Dashboard Visibility

System health information is continuously available for diagnostics.

Section 4 – Hardware Overview

4.1 Hardware Architecture

The weather station hardware is divided into:

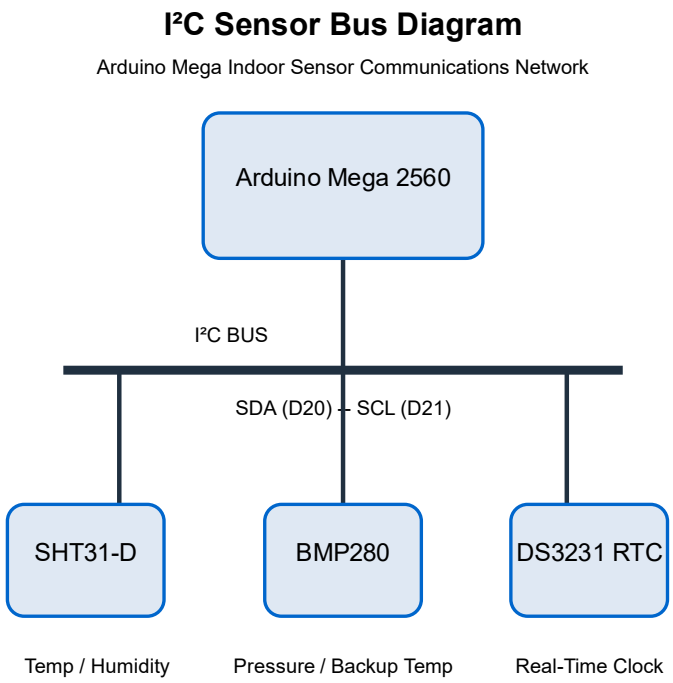
Indoor Hardware

Outdoor Hardware

Power System

Communications System

Figure 4-1. I²C Sensor Bus Diagram



Indoor sensor communications bus showing the Arduino Mega,

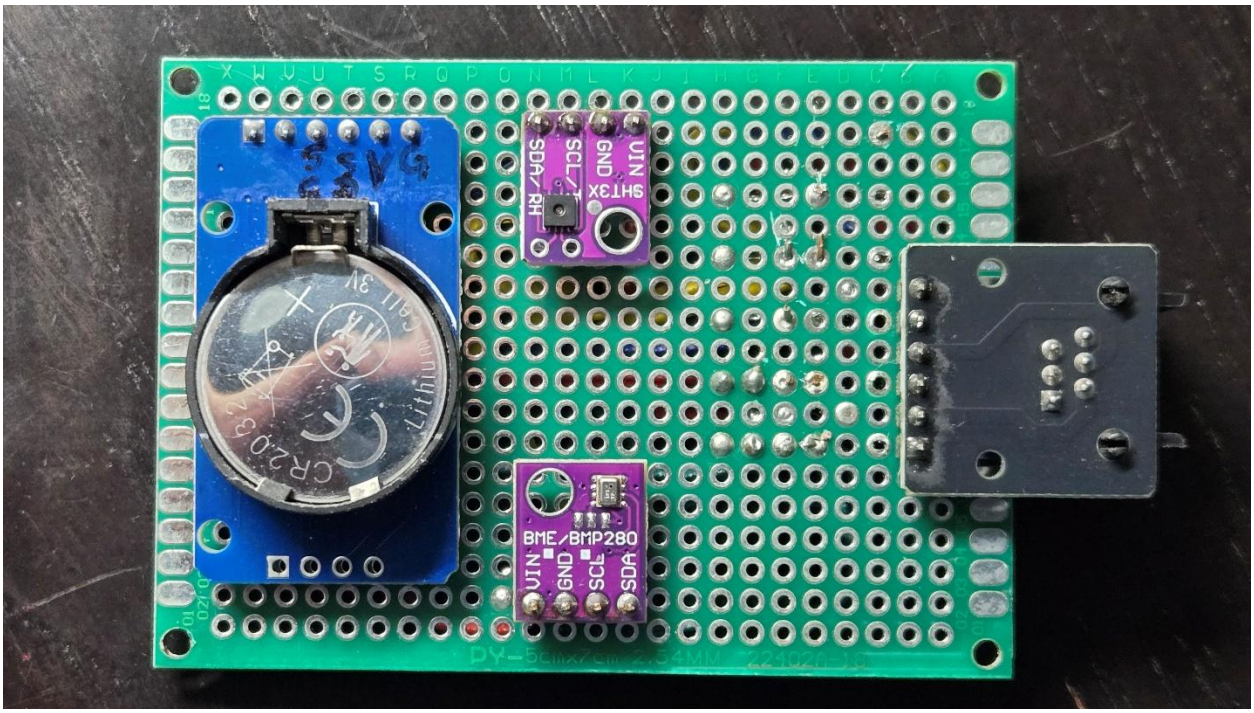
SHT31-D temperature/humidity sensor, BMP280 pressure sensor,

and DS3231 real-time clock connected through the shared I²C

interface using SDA (D20) and SCL (D21).

The SHT31-D, BMP280, and DS3231 RTC sensors are located on a small perfboard that is shielded from direct sun inside a radiation shield. Here is a picture of an earlier version of the perfboard:

Figure 4-2. Sensors inside Radiation Shield



The DS3231 RTC is on the left, showing its CR2032 cell. The SHT31-D sensor, which serves as the current primary temperature and humidity sensor, is the small purple module at top middle. The BMP280 pressure sensor is the small purple module at bottom middle of the board. The RJ12 socket for the cable connector is at the right.

Here is a picture of the underside of the perfboard, proving that it doesn't need to be beautiful in order to work:

Figure 4-3. Underside of Radiation Shield Perfboard

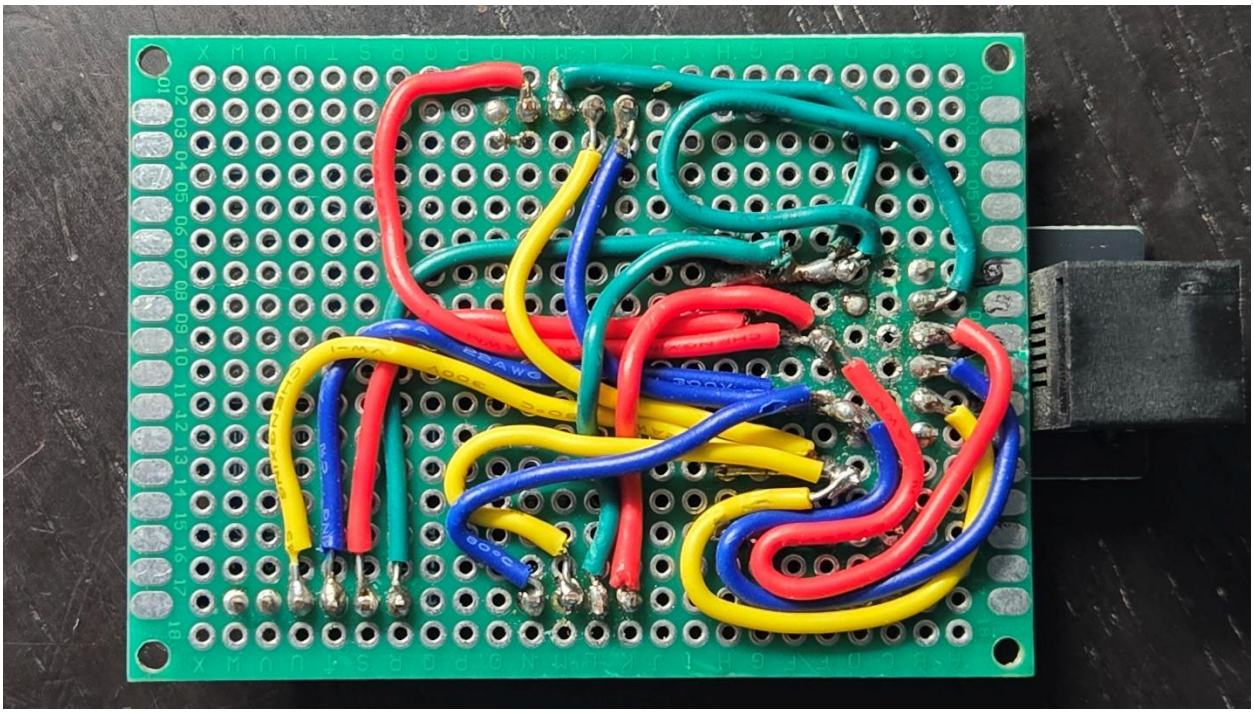


Figure 4-4. Radiation Shield Installation



Radiation shield outside office window, used for temperature and humidity measurements.

4.2 Indoor Hardware Inventory

Component	Function
Arduino Mega 2560	Main controller
LCD Keypad Shield	Local interface
SHT31-D	Temperature/Humidity
BMP280	Pressure
DS3231 RTC	Timekeeping
RYLR890	LoRa communications
ESP32-S3	Gateway controller

Figure 4-5. Indoor Electronics Diagram

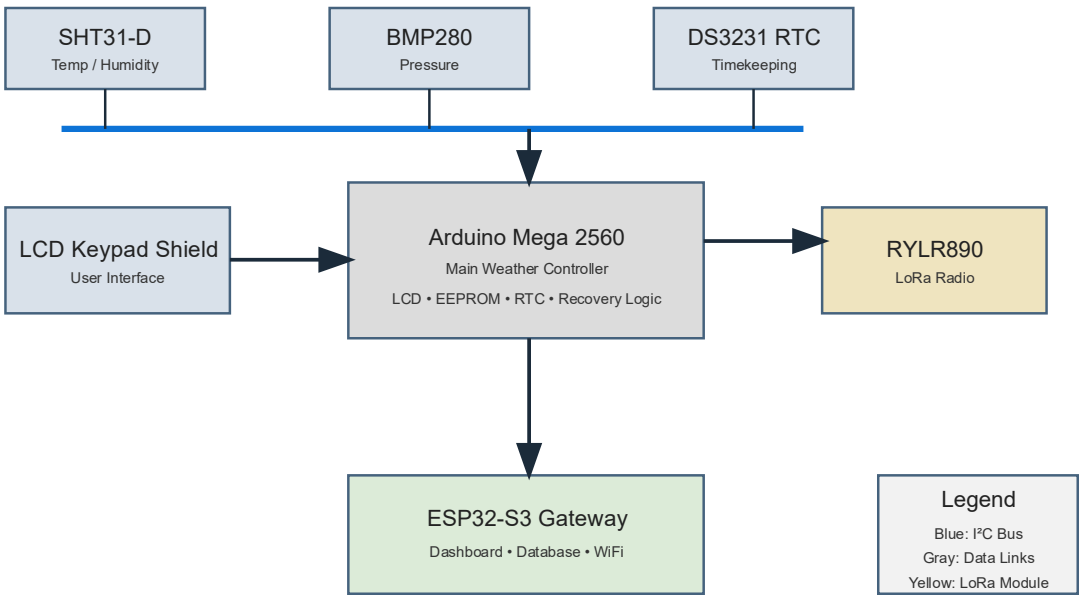
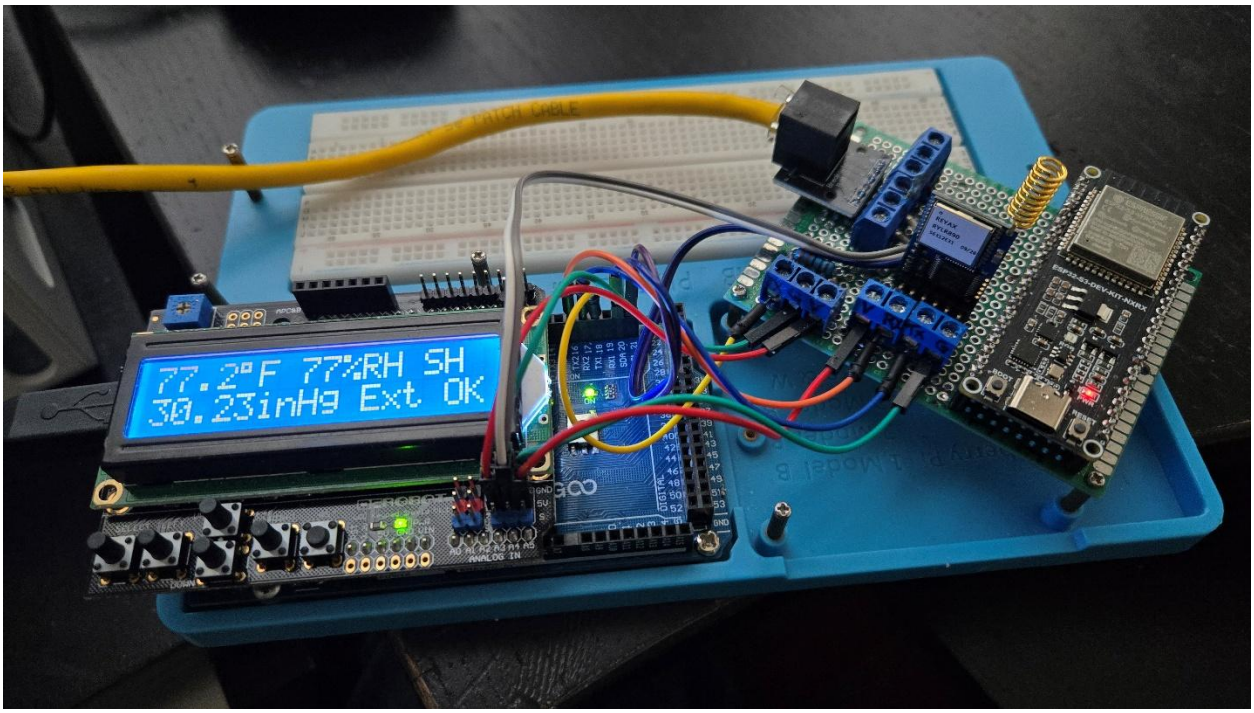


Figure 4-6. Indoor Controller Assembly



Indoor electronics assembly showing Arduino Mega, LCD hat, yellow cable to sensors outside the office window, LoRa transceiver, and ESP32-S3 gateway.

4.3 Outdoor Hardware Inventory

Component	Function
ESP32-C3	Outdoor controller
Wind Vane	Direction sensing
Anemometer	Wind-speed sensing
Rain Gauge	Rainfall measurement
RYLR890	LoRa communications
Antenna	RF communications

Figure 4-7. Outdoor Electronics Diagram

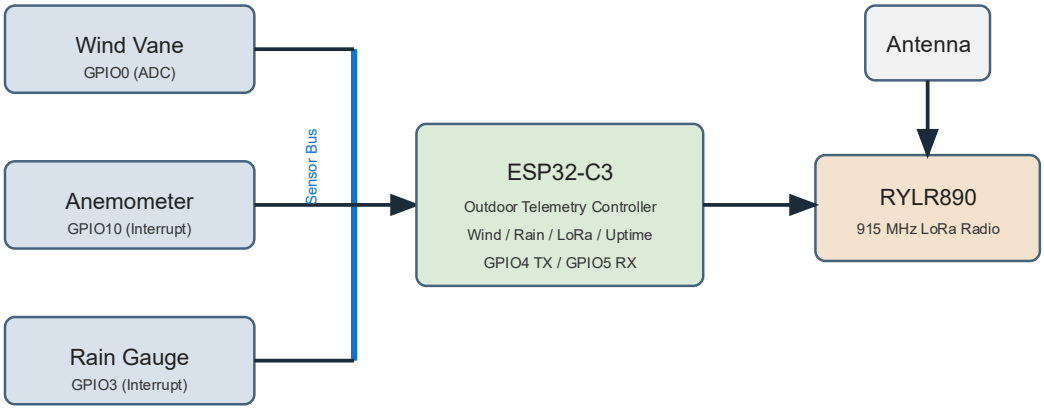


Figure 4-8. Outdoor Sensor Mast



Outdoor sensor mast showing weather sensors and supporting hardware.

4.4 Arduino Mega GPIO Assignments

Pin	Function
A0	**LCD keypad input**
D10	**LCD backlight**
D16	**LoRa TX**
D17	**LoRa RX**
D18	**UART1 TX to ESP32-S3**
D19	**UART1 RX from ESP32-S3**
D20	**SDA**
D21	**SCL**

UART Interface

The Arduino Mega communicates with the ESP32-S3 gateway using the hardware serial port **Serial1** operating at 9600 baud. The interface uses Arduino Mega pin **D18 (TX1)** to transmit weather and status data to the ESP32-S3 and **D19 (RX1)** to receive data from the gateway. On the ESP32-S3, the corresponding UART1 pins are **GPIO18 (RX)** and **GPIO17 (TX)**. This dedicated serial link provides the communications path between the weather-processing controller and the gateway subsystem.

4.5 ESP32-S3 GPIO Assignments

GPIO	Function
GPIO18	UART RX
GPIO17	UART TX

4.6 ESP32-C3 GPIO Assignments

GPIO	Function
GPIO10	Anemometer
GPIO3	Rain Gauge
GPIO0	Wind Vane ADC
GPIO4	LoRa RX
GPIO5	LoRa TX

4.7 Sensor Wiring Summary

Wind Vane

Connection	Signal
Pin 2	Ground
Pin 3	3.3V
Pin 5	ADC

Anemometer

Connection	Signal
Pin 3	GPIO10
Pin 4	Ground

Rain Gauge

Connection	Signal
Pin 3	GPIO3
Pin 4	Ground

Signal Conditioning Components

Signal	ESP32-C3 Pin	Conditioning Circuit
Wind Vane	GPIO0 (ADC)	12K pull-up resistor to 3.3V
Rain Gauge	GPIO3	10K pull-up resistor to 3.3V
Anemometer	GPIO10	Direct digital input

A 12K pull-up resistor was selected to optimize ADC separation between vane resistor-network states.

Figure 4-9. Solar Power System

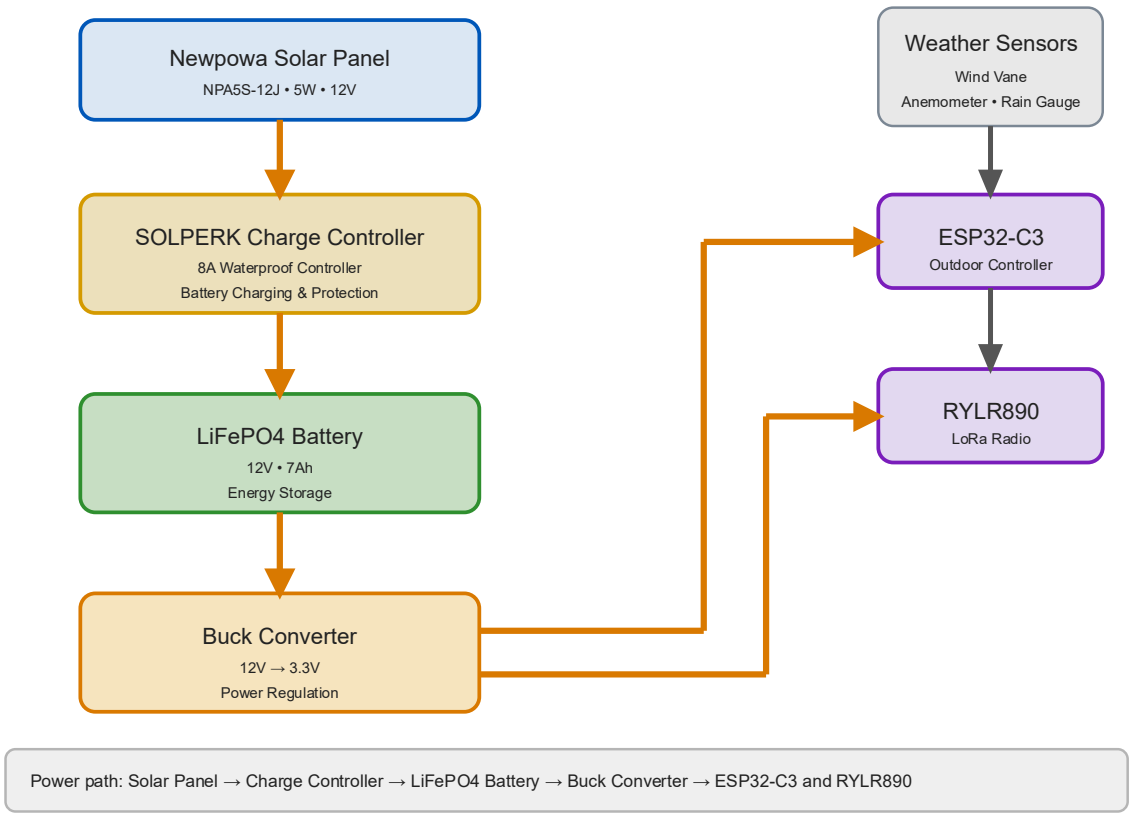


Figure 4-10. Outdoor Solar Power Assembly

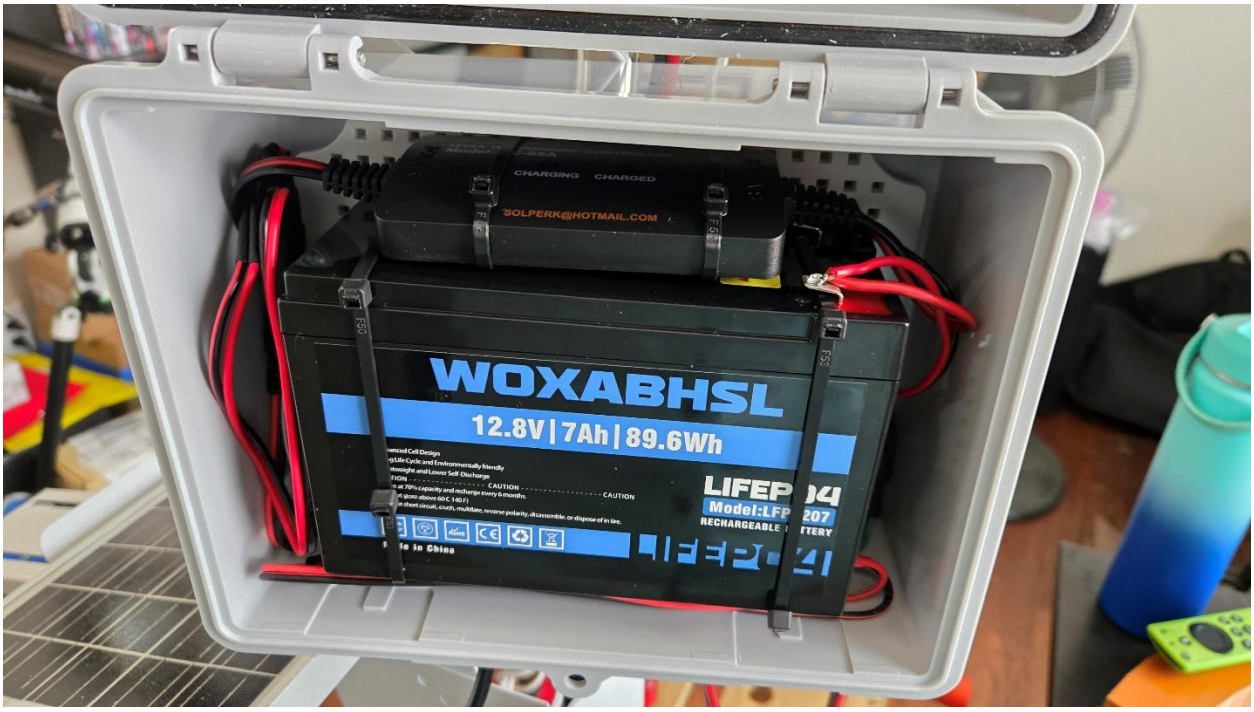
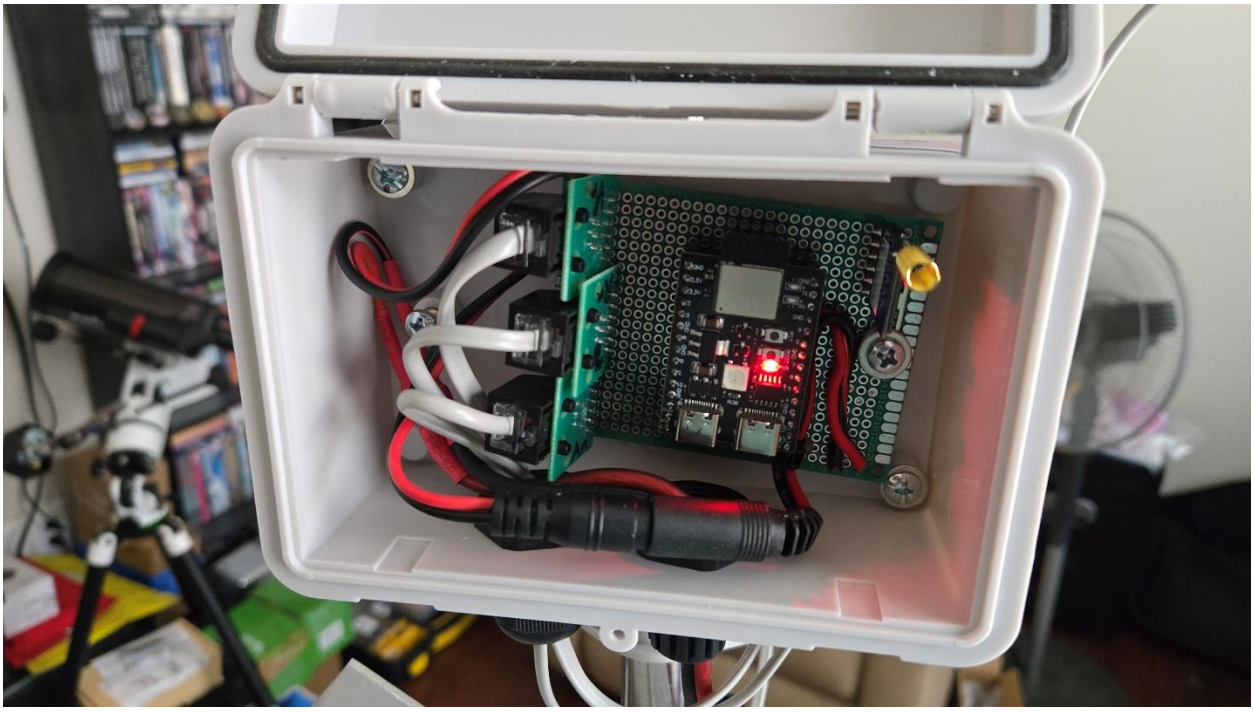


Figure 4-11. Outdoor Controller Electronics Assembly



These two photographs document the as-built implementation of the outdoor power and telemetry subsystems.

The solar power assembly contains the charge controller, LiFePO4 battery, wiring harnesses, and associated power-distribution components.

The controller electronics assembly contains sensor-interface wiring (RJ11 sockets), the ESP32-C3 outdoor controller, RYLR890 LoRa radio, buck converter, and signal-conditioning components.

These images provide a field-reference baseline for maintenance and troubleshooting.

(Configuration is shown as installed during Release 1.0 documentation.)

Table 4-7 Solar Hardware Inventory

Component	Manufacturer	Model	Specification	Purpose
Solar Panel	Newpowa	NPA5S-12J	5W, 12V Monocrystalline	Primary energy source
Charge Controller	SOLPERK	8A Controller	IP67 Rated	Battery charging and protection
Battery	WOXABHSL	LFP1207	12V 7Ah LiFePO4	Energy storage
Outdoor Controller	ESP32-C3	N/A	3.3V Controller	Outdoor telemetry processing
LoRa Radio	Reyax	RYLR890	915 MHz	Wireless communications
Antenna	Compatible 915 MHz	N/A	LoRa antenna	Radio communications

Design Objectives

The solar-power subsystem was selected to provide:

- Continuous outdoor operation
- No external power wiring
- Flexible sensor placement
- Improved installation options
- Reduced maintenance requirements

The battery and charge controller are sized to maintain operation during periods of reduced solar availability while supporting continuous weather monitoring and LoRa communications.

4.9 Communications Hardware

The communications subsystem consists of:

- LoRa indoor module
- LoRa outdoor module
- LoRa antennas
- UART interface
- Wi-Fi interface

These components provide all data transport throughout the system.

Sections 5–8

Firmware Architecture and Communications

Section 5 – Arduino Mega Firmware Architecture

5.1 Purpose

The Arduino Mega serves as the central supervisory controller for the weather station. While the outdoor ESP32-C3 acquires weather measurements and the ESP32-S3 provides network and dashboard services, the Mega functions as the operational core of the system.

Responsibilities include:

- User-interface management
- Environmental sensor processing
- RTC management
- EEPROM management
- Min/max record tracking
- Outdoor-node supervision
- Weather-data aggregation
- Communications coordination
- Fault-recovery management

The Mega is therefore the primary authority for weather-system state and operational control.

5.2 Software Architecture

The firmware utilizes a cooperative multitasking model built around the Arduino loop() function.

Rather than relying on a real-time operating system, the firmware schedules periodic activities through millisecond timing intervals using:

C++

millis()

This approach allows multiple independent activities to operate concurrently without blocking user interaction.

Major tasks include:

Sensor Acquisition

LCD Updates

Button Monitoring

LoRa Reception

ESP32 Packet Generation

EEPROM Management

Min/Max Tracking

Recovery Management

Diagnostics

5.3 User Interface Architecture

LCD Hardware

The user interface utilizes a standard LCD Keypad Shield mounted directly on the Arduino Mega.

The shield provides:

- 16x2 LCD display
- Five navigation buttons
- Backlight control

Buttons are connected through a resistor ladder network monitored on analog input A0.

Button Functions

Button	Function

RIGHT	Navigate Forward
LEFT	Navigate Back
UP	Value Increase / Menu Navigation
DOWN	Value Decrease / Menu Navigation
SELECT	Accept / Enter

Menu Structure

The firmware implements a hierarchical menu system.

Primary screens:

Super Screen

Wind Screen

Rain Screen

Clock Screen

Pressure Screen

Min Temperature

Max Temperature

Humidity Records

Pressure Records

Settings

Diagnostics

Button Test

The menu system provides complete access to operational and diagnostic functions without requiring network connectivity.

5.4 Sensor Acquisition Engine

SHT31-D

Primary measurements:

- Temperature
- Relative Humidity

The SHT31-D serves as the preferred environmental sensor because it provides both measurements from a single source.

BMP280

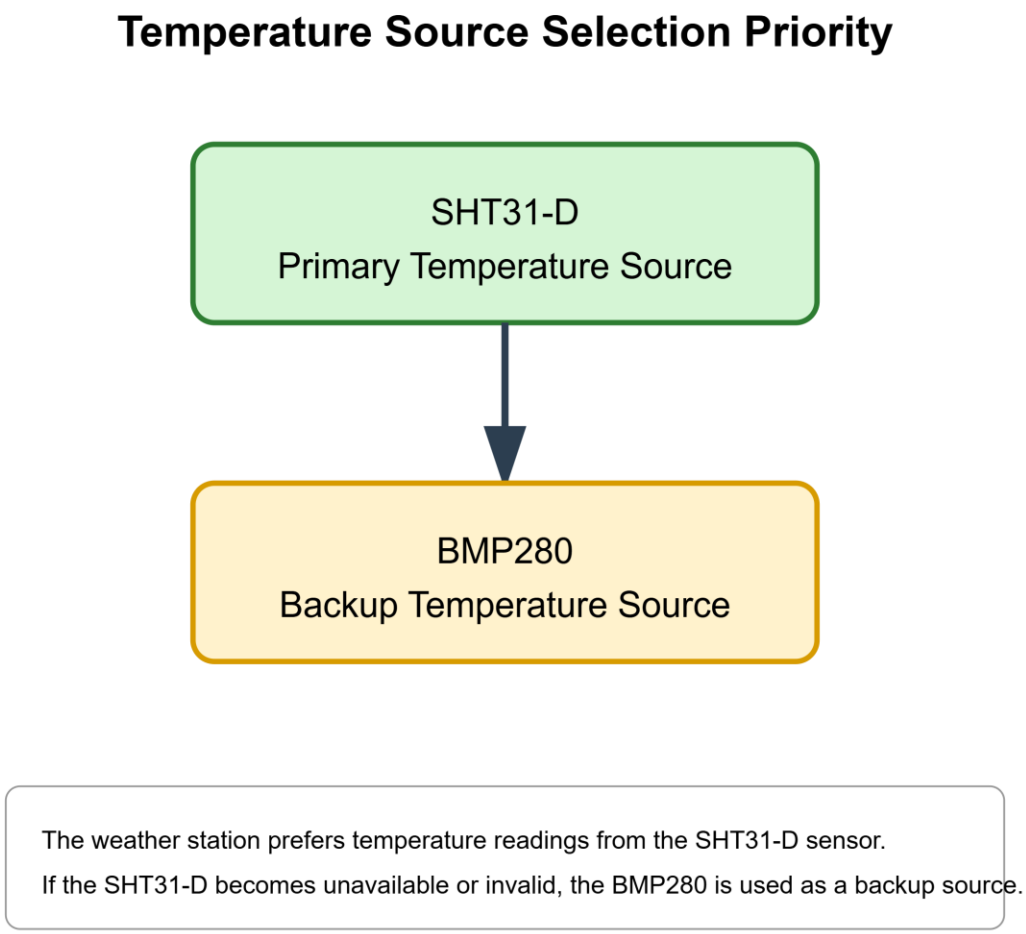
Measurements:

- Atmospheric Pressure
- Backup Temperature

The BMP280 primarily supplies pressure measurements but also provides secondary temperature data.

Sensor Redundancy

Temperature processing includes source-selection logic.



If the primary source becomes invalid, the firmware automatically transitions to the backup source.

This improves system resilience and reduces data loss.

5.5 Min/Max Tracking Engine

The weather station continuously tracks environmental extremes.

Tracked values:

Minimum Temperature

Maximum Temperature

Minimum Humidity

Maximum Humidity

Minimum Pressure

Maximum Pressure

Associated timestamps are stored for every record.

Timestamp Components

Records contain:

Month

Day

Hour

Minute

allowing operators to determine when the extreme occurred.

5.6 Temperature Suppression System

Objective

The suppression system compensates for known afternoon temperature influences that can elevate readings beyond representative ambient conditions.

Operating Window

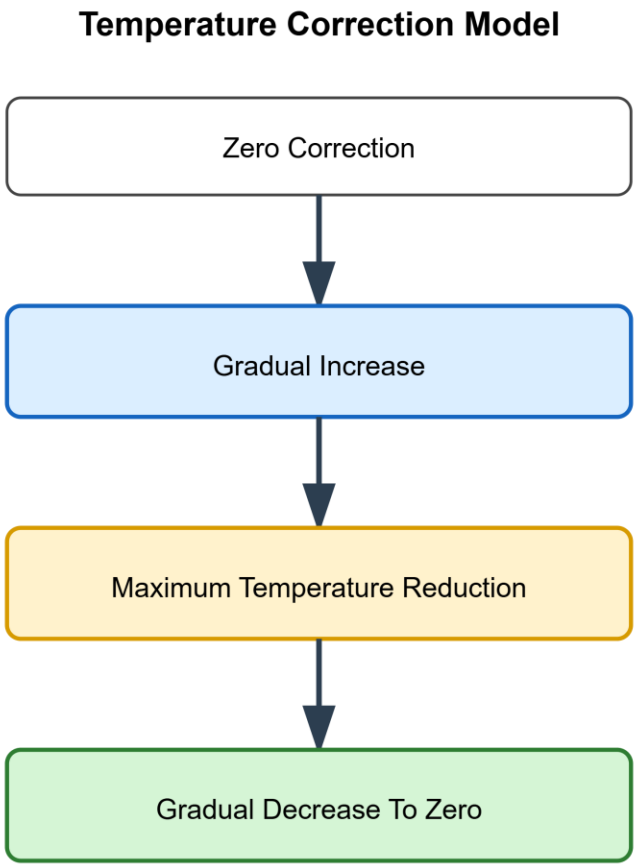
The correction operates during:

15:00 to 20:00

Correction Model

The adjustment follows a smooth sinusoidal curve.

Conceptually:



This avoids abrupt transitions while maintaining consistent environmental reporting.

5.7 EEPROM Manager

The firmware stores persistent data in EEPROM.

Stored information includes:

Min Temperature

Max Temperature

Min Humidity

Max Humidity

Min Pressure

Max Pressure

Altitude

Reset Information

EEPROM preservation ensures critical information survives power interruptions.

5.8 RTC Management

The DS3231 RTC provides:

- Date
- Time
- Daily rollover detection

The RTC acts as the authoritative time source for:

Weather Records

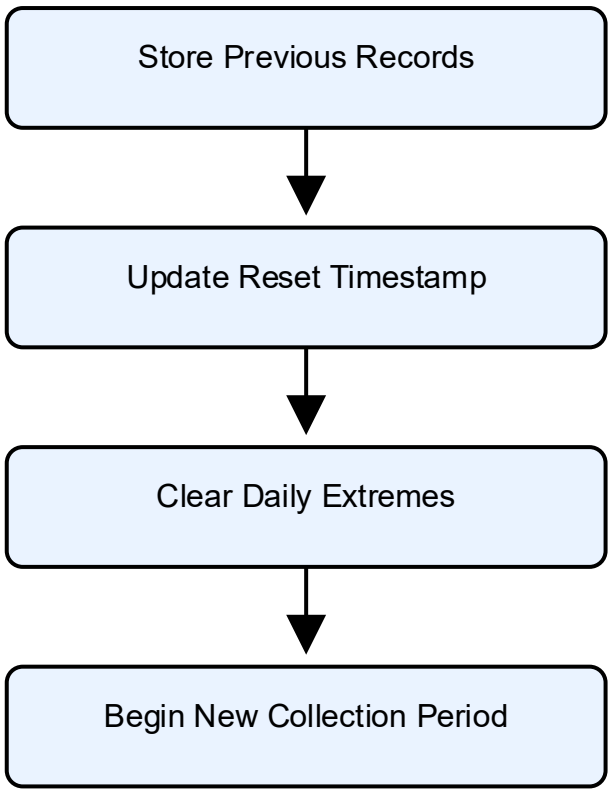
Database Uploads

Min/Max Timestamps

Daily Resets

5.9 Daily Reset Processing

At day rollover:



This process ensures clean separation between daily observation periods.

5.10 Outdoor Communications Supervision

The Mega continuously evaluates outdoor telemetry health.

Monitoring criteria include:

Packet Arrival

Packet Freshness

Outdoor Uptime

Recovery State

Warning Threshold

30 Seconds

without valid updates.

Recovery Threshold

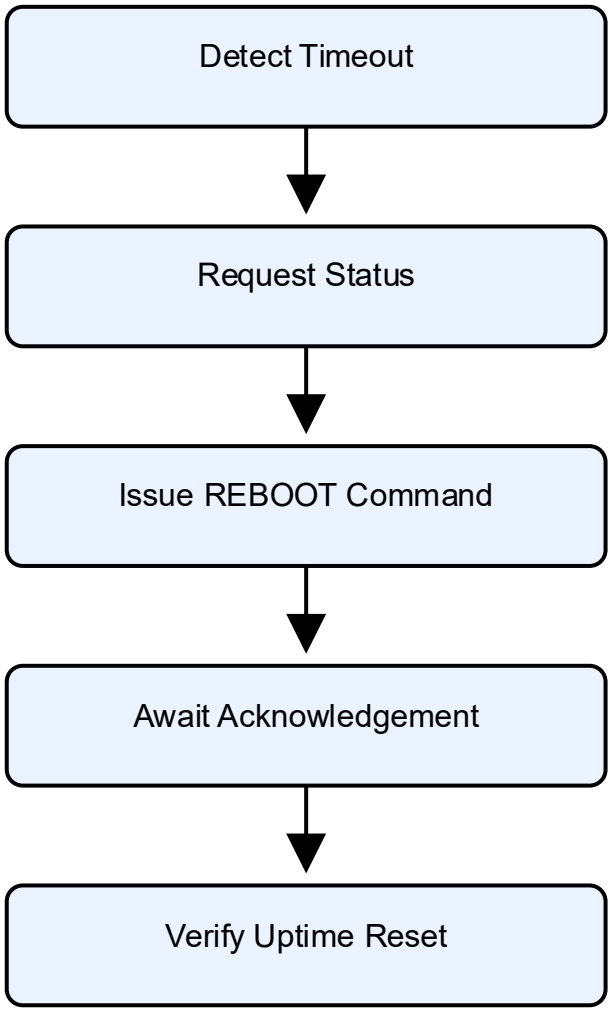
60 Seconds

without valid updates.

5.11 Remote Reboot Manager

If communications are lost, the Mega can initiate recovery.

Process:



Maximum retries: 3

5.12 Weather Packet Generation

Once all measurements are consolidated, the Mega generates a weather-report packet for the ESP32-S3.

Typical contents include:

Temperature

Humidity

Dew Point

Pressure

Wind Speed

Wind Gust

Wind Direction

Rainfall

Rain Rate

Daily Rain

Outdoor Uptime

Date

Time

Transmission interval: Approximately 5 seconds

Section 6 – ESP32-S3 Gateway Firmware

6.1 Purpose

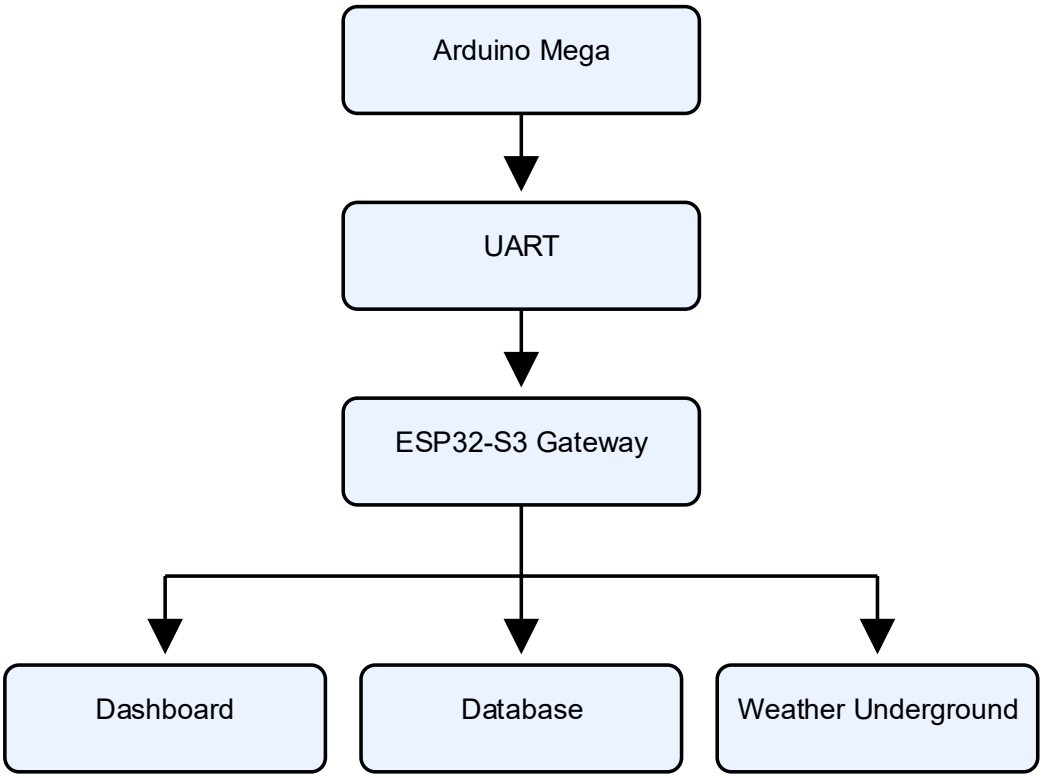
The ESP32-S3 acts as the Internet gateway and data-presentation platform.

Responsibilities include:

- Wi-Fi connectivity

- Dashboard generation
- Database uploads
- Weather Underground uploads
- Status monitoring
- Communications supervision

6.2 Gateway Data Flow



The gateway does not generate weather data.

Instead, it receives processed information from the Mega.

6.3 UART Processing Engine

The ESP32-S3 continuously monitors UART traffic from the Mega.

The parser extracts:

Temperature

Humidity

Pressure

Wind Data

Rain Data

System Status

These values become dashboard variables.

6.4 Dashboard Engine

The gateway hosts a browser-accessible dashboard.

Displayed information includes:

Temperature

Humidity

Dew Point

Pressure

Wind

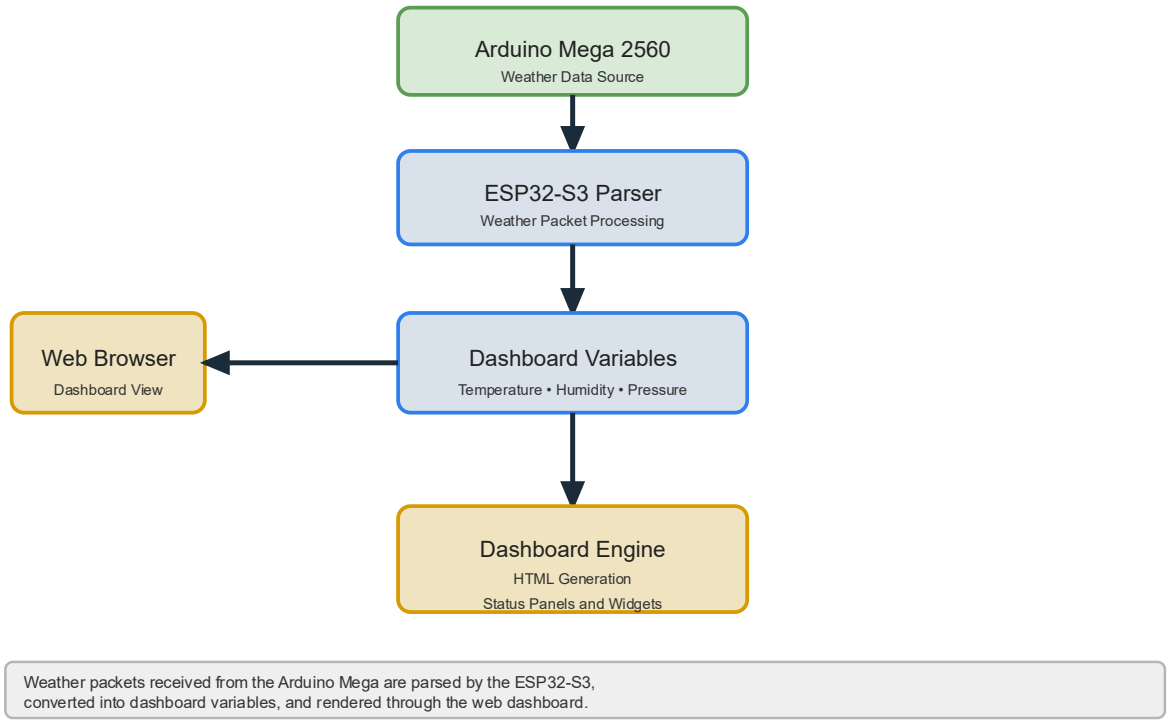
Rain

Outdoor Uptime

Weather Underground Status

The dashboard serves as the primary remote-monitoring interface.

Figure 6-1. Dashboard Data Flow



Processing path used by the ESP32-S3 gateway to transform weather packets into dashboard content.

6.5 Weather Underground Integration

The ESP32-S3 performs all Weather Underground communications.

Functions:

Generate Upload Request

Transmit Data

Evaluate Success

Update Status

Dashboard classifications:

ONLINE

DEGRADED

OFFLINE

6.6 Database Upload Engine

The gateway uploads weather information to:

weatherdb.php

Data categories include:

- Environmental values
- Health metrics
- Upload status
- Time information

6.7 Moisture Analysis

The gateway calculates:

Dew Potential

Moisture Classification

These calculated values simplify interpretation of humidity data.

6.8 Health Monitoring

The gateway tracks:

Wi-Fi Status

Database Status

Weather Underground Status

Upload Success

These values provide early warning of communications issues.

6.9 Recovery Logic

If communications failures exceed acceptable limits, the ESP32-S3 may initiate self-recovery procedures.

Goals:

- Restore uploads
- Restore dashboard operation
- Restore connectivity

Section 7 – ESP32-C3 Outdoor Node Firmware

7.1 Purpose

The outdoor ESP32-C3 functions as the environmental telemetry controller.

Responsibilities include:

- Wind-speed measurement
- Wind-direction processing
- Rainfall measurement
- LoRa communications
- Uptime reporting

7.2 Sensor Interfaces

Wind Vane

GPIO0

ADC-based sensing.

Anemometer

GPIO10

Interrupt-driven pulse counting.

Rain Gauge

GPIO3

Interrupt-driven pulse counting.

7.3 Wind Measurement System

The anemometer generates reed-switch transitions.

Process:

Pulse Detection

↓

Count Accumulation

↓

Wind-Speed Calculation

7.4 Gust Processing

The firmware continuously tracks peak instantaneous wind activity.

Result:

Maximum Gust

reported within outgoing telemetry packets.

7.5 Wind-Direction Processing

The wind vane produces a variable resistance network.

The firmware converts:

ADC Reading

into

Compass Direction

in

Degrees

before transmission.

7.6 Rainfall Processing

The tipping-bucket rain gauge generates pulse events.

Measurements include:

Rain Tips

Rain Rate

Daily Rainfall

7.7 LoRa Packet Construction

The outdoor node builds telemetry packets approximately every five seconds.

Packet contents:

Wind Speed

Wind Gust

Wind Direction

Rain Rate

Daily Rain

Rain Tips

Uptime

7.8 Uptime Monitoring

The ESP32-C3 continuously reports uptime.

Benefits:

- Health monitoring
- Recovery verification
- Communications diagnostics

7.9 Remote Command Processing

Supported commands:

STATUS

REBOOT

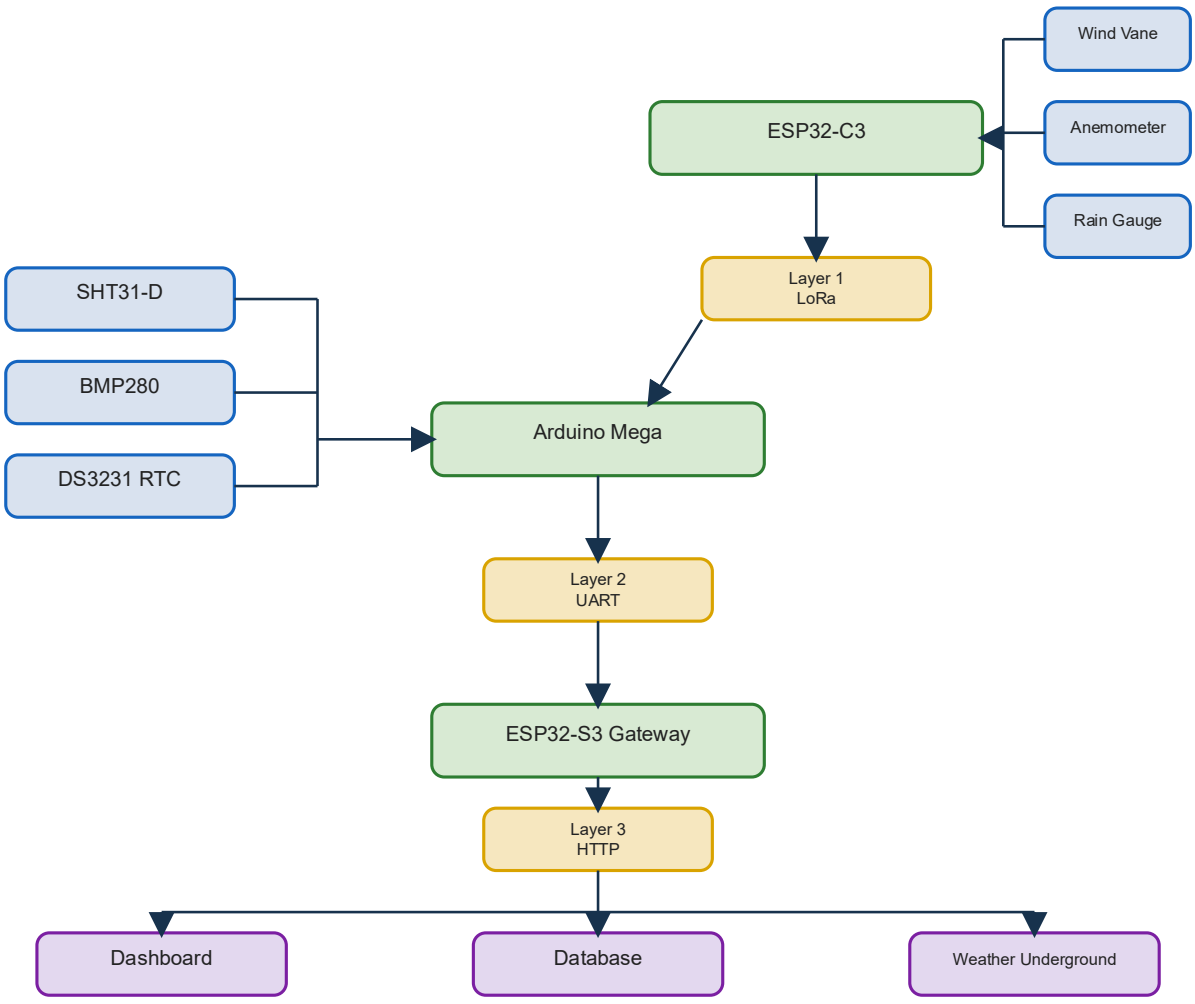
These commands allow remote supervision by the Arduino Mega.

Section 8 – Communications Protocols

8.1 Purpose

The communications subsystem links all weather-station components into a unified architecture.

Figure 8-1. Communications and Data Acquisition Architecture



End-to-end weather-data flow from outdoor sensors through the control system and into dashboard and cloud services.

8.2 Communications Layers

Layer 1

LoRa

Outdoor telemetry.

Layer 2

UART

Mega-to-gateway communications.

Layer 3

HTTP

Internet communications.

8.3 LoRa Weather Packet

Packet identifier:

<OUT>

Structure:

WindSpeed,

WindGust,

Direction,

RainRate,

DailyRain,

RainTips,

Uptime

Update interval: ~5 Seconds

8.4 Status Protocol

Query:

<STATUS?>

Response:

<STATUS>

Purpose:

Verify node health when telemetry becomes stale.

8.5 Remote Reboot Protocol

Command:

<REBOOT>

Acknowledgement:

<REBOOTING>

Verification:

Uptime Reset

8.6 UART Weather Packet

The Mega forwards consolidated weather information to the ESP32-S3.

Fields include:

Temperature

Humidity

Pressure

Wind

Rain

Uptime

Date

Time

8.7 HTTP Uploads

Destination services:

Database

Weather Underground

The ESP32-S3 performs all upload operations.

8.8 Communications Timing

Activity	**Interval**
LoRa Telemetry	~5 sec
UART Update	~5 sec
Status Check	~30 sec
Reboot Trigger	~60 sec

Retry Count	3

8.9 Communications Health Monitoring

The architecture continuously monitors:

LoRa Health

Gateway Status

Upload Status

Database Connectivity

Weather Underground Connectivity

This layered monitoring structure provides robust fault detection and recovery.

Sections 9–14

Database, Dashboard, Algorithms, Installation, Troubleshooting, and Maintenance

---

Section 9 – Database Design

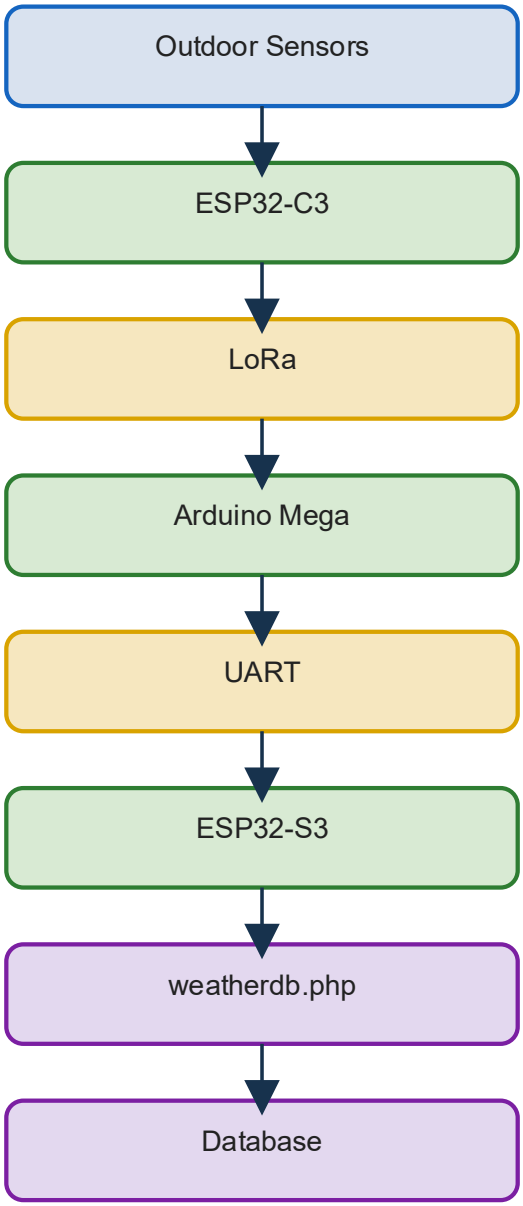
9.1 Purpose

The database subsystem provides long-term storage of weather and operational data generated by the weather station. While the Arduino Mega and ESP32-S3 maintain current operating values, the database preserves historical observations for reporting, troubleshooting, trend analysis, and future dashboard enhancements.

The database serves as the permanent historical record of station operation.

9.2 Database Architecture

The database resides downstream of the weather-acquisition chain.



This separation allows weather acquisition to continue even when database services are unavailable.

9.3 Design Objectives

The database design was developed to meet several goals:

- Provide historical weather storage
- Preserve raw and corrected values
- Support future reporting tools
- Support dashboard displays
- Simplify troubleshooting
- Record system-health information

9.4 Observation Records

Each database record represents a single observation event.

Typical categories include:

Environmental Conditions

System Health

Upload Status

Timestamp Information

Each observation is timestamped and archived for later retrieval.

9.5 Environmental Data

Stored environmental information includes:

Temperature

temp

rawtemp

adjfactor

Humidity

humidity

dewpt

moist

dewpot

Pressure

presshpa

pressinhg

Wind

winddir

windspd

windgust

windlast

Rain

rainrate

rainday

raintips

9.6 System-Health Data

The database stores operational status information.

Examples:

Outdoor Uptime

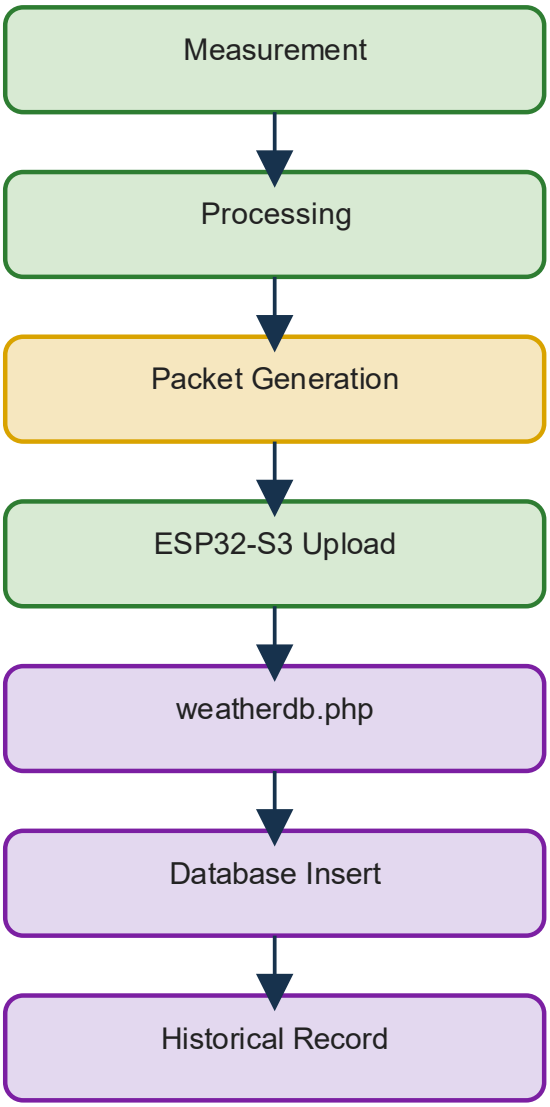
Weather Underground Status

Last Connection

Next Upload Information

These values greatly simplify fault diagnosis.

9.7 Observation Lifecycle



9.8 Database Failure Tolerance

One design objective was fault isolation.

If database services fail:

Weather Collection Continues

LCD Continues

LoRa Continues

Dashboard Continues

Only historical record insertion is affected.

9.9 Future Database Applications

The current schema supports:

- Historical weather charts
- Rainfall reports
- Uptime analytics
- Reliability analysis
- Trend forecasting

without requiring major structural changes.

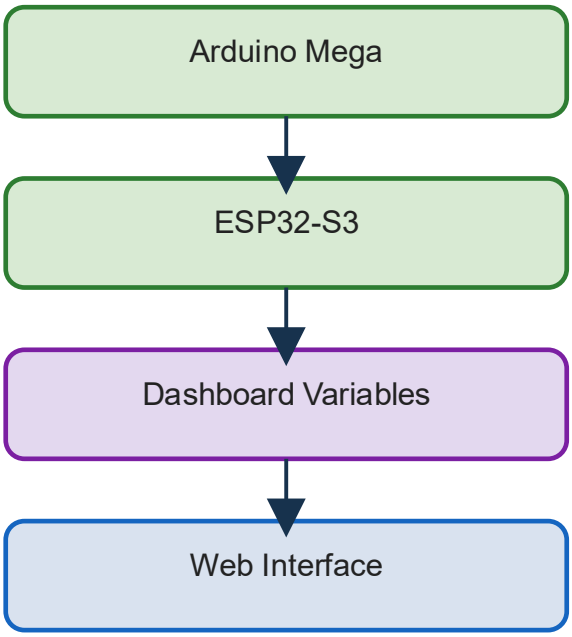
Section 10 – Dashboard Design

10.1 Purpose

The dashboard provides remote visibility into weather conditions and station health.

Unlike the LCD display, which is intended for local operation, the dashboard is accessible through a standard web browser and provides a complete system overview.

10.2 Dashboard Architecture



The dashboard reflects the most recent weather information received by the gateway.

10.3 Design Goals

The dashboard was designed to provide:

- Rapid readability
- Mobile compatibility
- System-health visibility
- Weather Underground visibility
- Near real-time updates

10.4 Main Dashboard Components

The dashboard displays:

Temperature

Humidity

Dew Point

Pressure

Wind

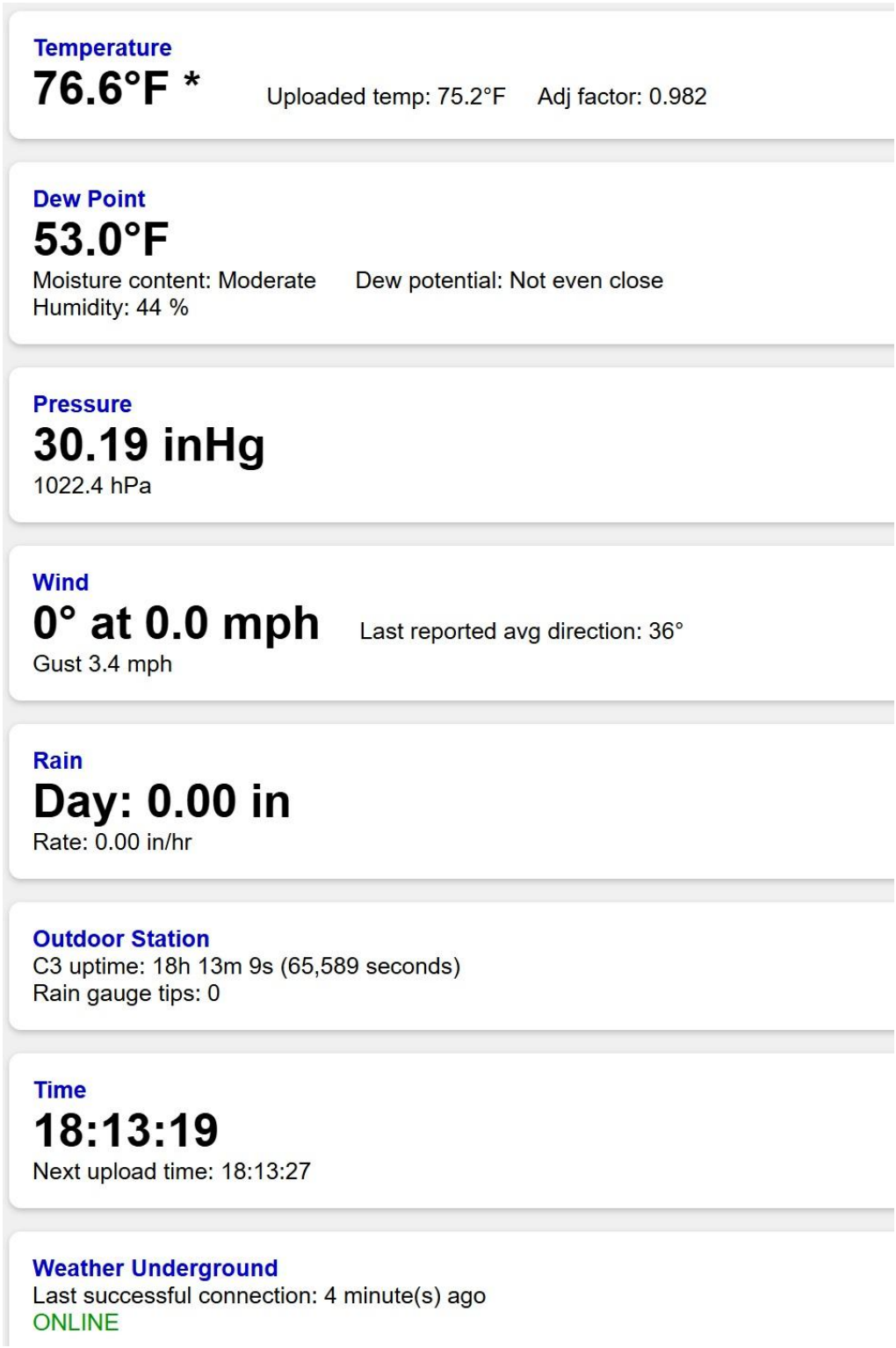
Rainfall

Outdoor Uptime

Weather Underground Status

Date/Time

Figure 10-1. Dashboard Screenshot



Primary localhost dashboard interface showing current weather and system status.

10.5 Temperature Display

The temperature section shows:

- Corrected temperature
- Sensor-derived values
- Suppression-adjusted values

The displayed value represents the station's preferred weather temperature.

10.6 Humidity and Dew Point

The humidity section displays:

Relative Humidity

Dew Point

Moisture Classification

providing a quick understanding of atmospheric conditions.

10.7 Pressure Display

Pressure information is displayed using corrected sea-level pressure.

Available units include:

hPa

inHg

10.8 Wind Display

Wind information includes:

Current Speed

Wind Gust

Direction

These values originate from the outdoor node.

10.9 Rainfall Display

The rainfall section provides:

Rain Rate

Daily Rainfall

Rain-Gauge Status

This allows rapid assessment of precipitation conditions.

10.10 Outdoor Node Status

The dashboard displays:

Outdoor Uptime

Communication Health

A rapidly growing uptime indicates healthy operation.

10.11 Weather Underground Status

Status indicators:

ONLINE

DEGRADED

OFFLINE

provide visibility into cloud-upload performance.

10.12 Future Dashboard Enhancements

Potential upgrades include:

- Historical charts
- Rainfall graphs
- Pressure trends
- Wind roses
- Battery telemetry

Section 11 – Environmental Algorithms

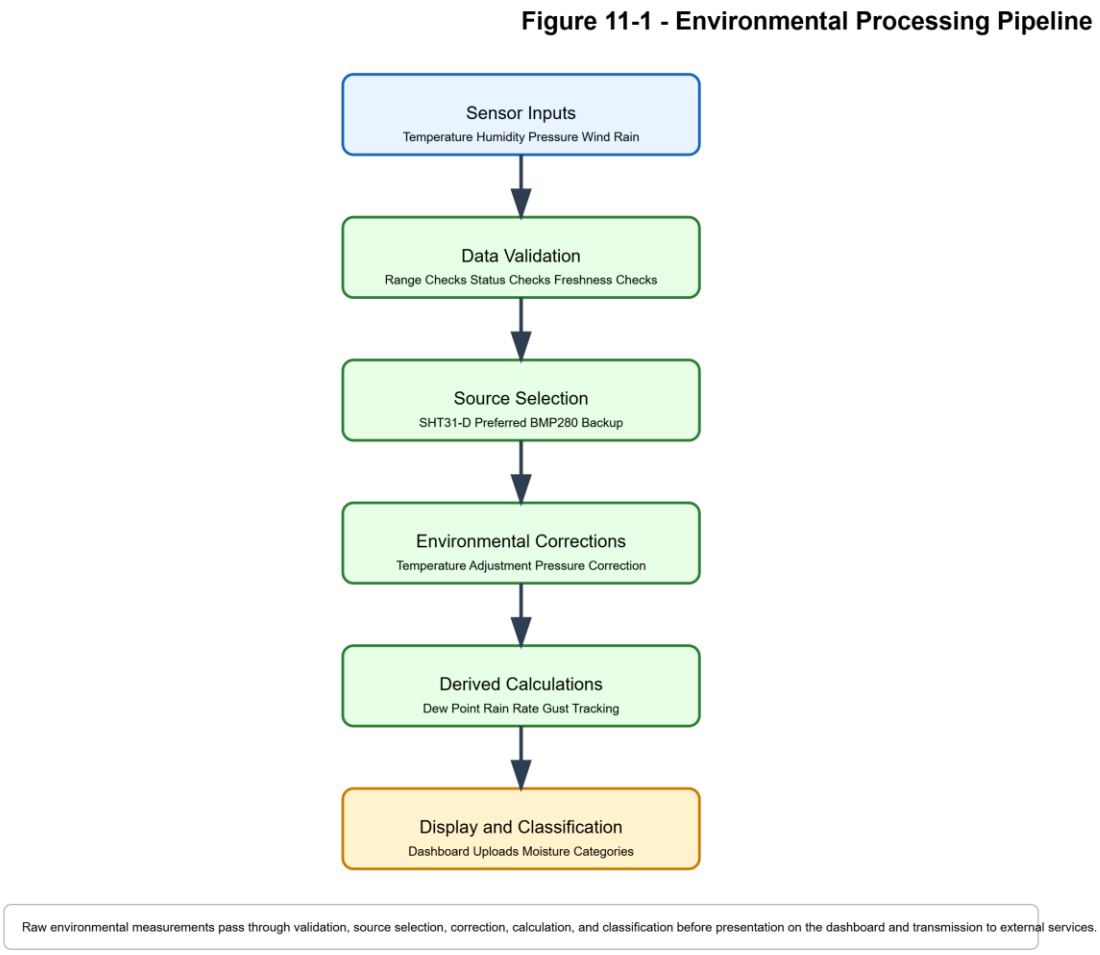
11.1 Purpose

Environmental algorithms convert raw sensor measurements into useful meteorological information.

These algorithms improve usability while preserving scientific value.

11.2 Processing Pipeline

Figure 11-1. Environmental Processing Pipeline



Environmental-data processing sequence from raw sensor measurements through correction, calculation, classification, display, and upload.

11.3 Temperature Source Selection

Temperature may originate from:

SHT31-D

BMP280

The system automatically selects the preferred valid source.

11.4 Temperature Suppression Algorithm

The firmware contains a time-based correction system intended to minimize elevated readings during a known afternoon interval.

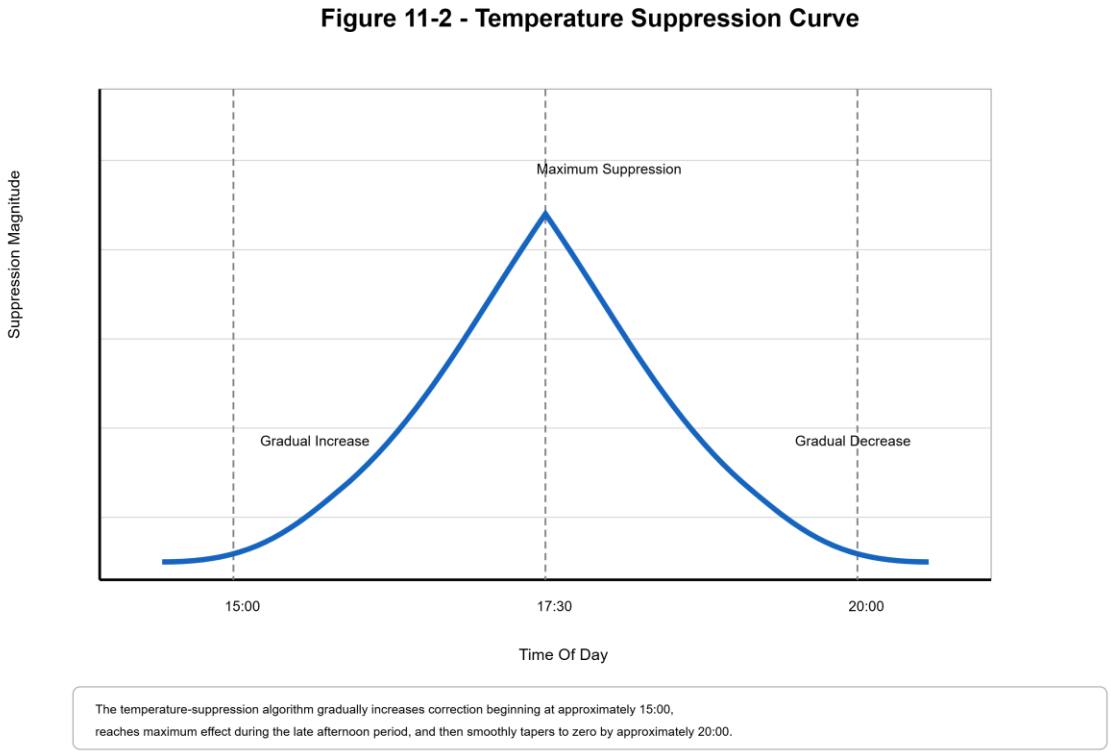
Operating Window

15:00 – 20:00

Characteristics

- Smooth transition
- Gradual increase
- Gradual decrease
- No abrupt step changes

Figure 11-2. Temperature Suppression Curve



Representative temperature-adjustment profile used to compensate for afternoon heating effects while avoiding abrupt transitions.

11.5 Dew-Point Calculation

Dew point is calculated using temperature and humidity.

General process:

Temperature

+

Humidity

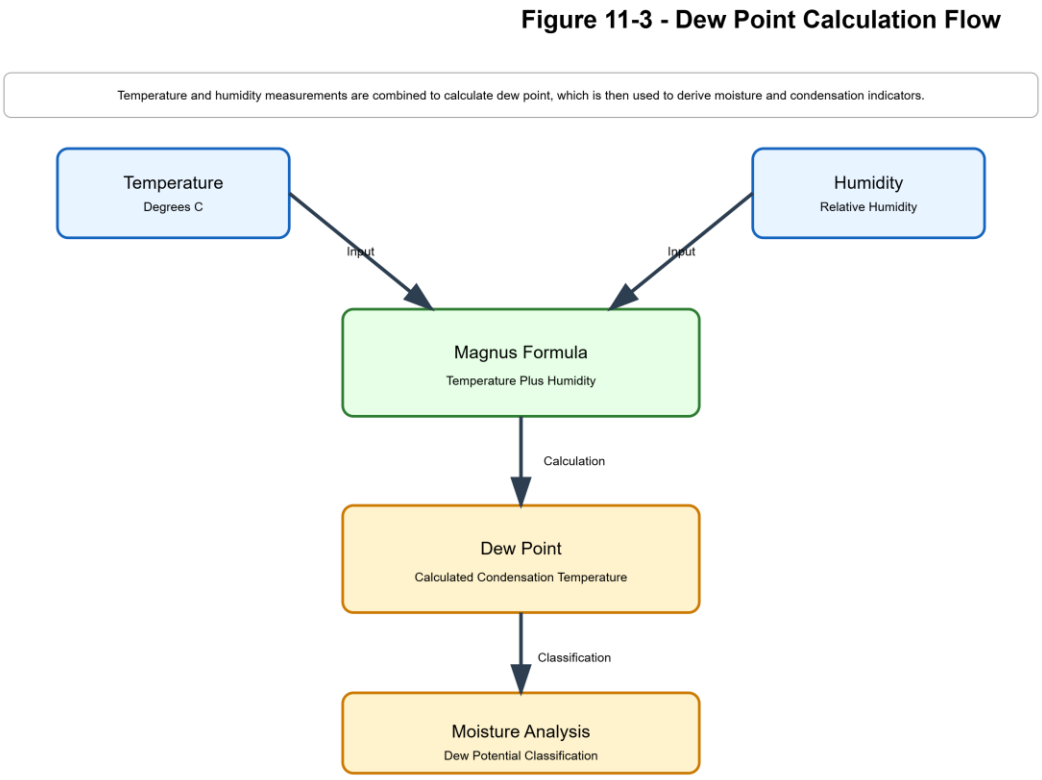
↓

Magnus Equation

↓

Dew Point

Figure 11-3. Dew Point Calculation Flow



Dew-point generation process utilizing temperature and humidity measurements.

11.6 Pressure Correction

Pressure readings are corrected using station altitude.

Inputs:

Observed Pressure

Altitude

Output:

Sea-Level Pressure

This produces weather values comparable to nearby reporting stations.

11.7 Wind Processing

The outdoor node performs:

Pulse Counting

Wind-Speed Calculation

Gust Tracking

Direction Processing

before transmission.

11.8 Rainfall Processing

Rainfall calculations are derived from tipping-bucket events.

Generated values:

Rain Tips

Rain Rate

Daily Rainfall

11.9 Moisture Classification

The system generates human-readable environmental classifications from weather conditions.

Examples may include:

Dry

Comfortable

Humid

Very Humid

Section 12 – Installation and Physical Deployment

12.1 Purpose

This section describes physical deployment of the complete weather station.

The installation process includes:

- Mechanical mounting
- Electronics installation
- Power-system installation
- Communications verification

12.2 Physical Assemblies

The station consists of:

Outdoor Sensor Mast

Outdoor Electronics Enclosure

Solar Power System

Indoor Controller

12.3 Site Selection

Ideal installation locations provide:

- Good wind exposure
- Open rainfall collection
- Reliable LoRa communications
- Sufficient solar exposure

Figure 12-1. Outdoor Sensor Mast



12.4 Wind Sensor Placement

Wind sensors should:

- Avoid obstructions
- Remain well above nearby structures
- Maintain unobstructed airflow

12.5 Rain Gauge Placement

The rain gauge should:

- Remain level
- Avoid roof runoff
- Avoid splash contamination
- Remain free of overhead obstructions

12.6 Radiation Shield Installation

The shield should:

- Remain ventilated
- Remain clean
- Avoid direct solar loading

Figure 12-2. Radiation Shield



12.7 Solar Power Installation

Install the solar power subsystem in accordance with Figure 4-7. Verify correct polarity between the solar panel, charge controller, battery, and buck converter before applying power.

12.8 Indoor Controller Installation

The indoor station should be installed:

- Indoors
- Within LoRa range
- Near power
- Accessible for maintenance

12.9 Commissioning Tests

Verify:

LoRa Communication

Dashboard Operation

Database Uploads

Weather Underground Uploads

Sensor Function

RTC Accuracy

before placing the station into service.

Section 13 – Troubleshooting and Diagnostics

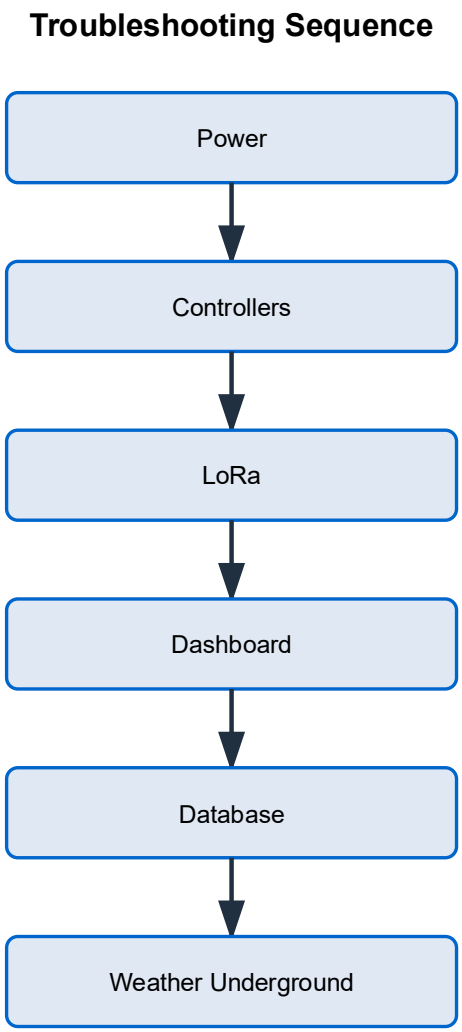
13.1 Purpose

The troubleshooting process is designed to rapidly isolate system failures.

The architecture intentionally exposes diagnostic information through:

- LCD displays
- Dashboard indicators
- Uptime reporting
- Communications status

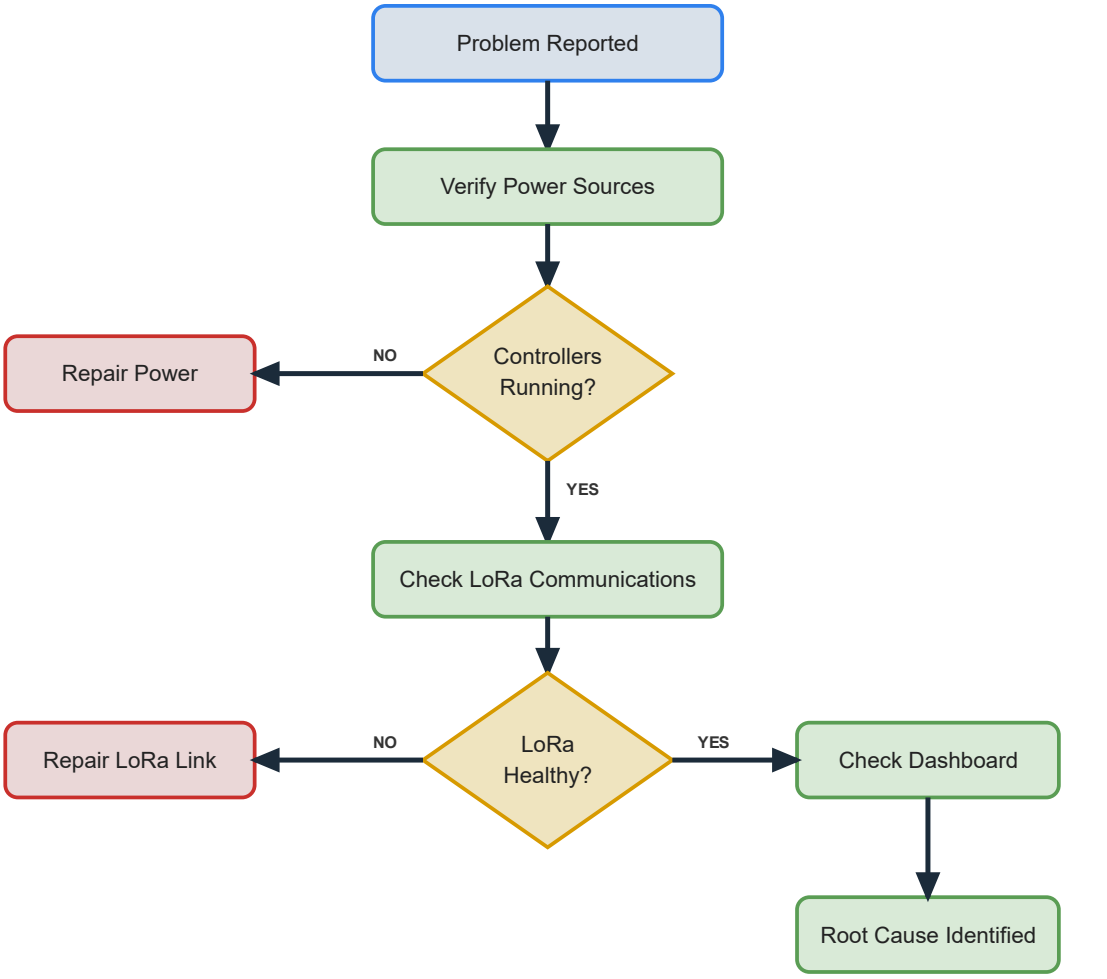
13.2 Troubleshooting Sequence



Recommended sequence

When diagnosing faults, begin with power verification and proceed sequentially through controllers, LoRa communications, dashboard operation, database services, and Weather Underground connectivity.

Figure 13-1. Troubleshooting Workflow



Follow the troubleshooting sequence from power verification through communications and dashboard validation to rapidly isolate the source of system failures.

Recommended fault-isolation process for weather-station troubleshooting.

13.3 Ext NO Condition

Indicates stale outdoor communications.

Possible causes:

- Outdoor-node outage
- LoRa failure
- Antenna issue
- Power-system failure

13.4 Wind Data Missing

Potential causes:

Sensor Failure

LoRa Failure

GPIO Wiring Problem

Inspection should begin with communications verification.

13.5 Rainfall Problems

Possible causes:

Debris

Bucket Binding

Wiring Failure

Pull-Up Resistor Failure

13.6 Incorrect Time

Potential causes:

RTC Configuration

RTC Battery

RTC Failure

13.7 Dashboard Not Updating

Possible causes:

UART Failure

Gateway Failure

Parser Failure

13.8 Weather Underground Offline

Potential causes:

Internet Outage

Wi-Fi Failure

DNS Failure

Credential Issue

13.9 Database Upload Failure

If dashboard updates continue but database records stop:

Likely Server-Side Failure

rather than weather acquisition failure.

13.10 Failure-Isolation Matrix

Symptom	Likely Cause
Ext NO	Outdoor Communication Failure
No Wind	Sensor / LoRa Problem
Frozen Dashboard	Gateway Problem
OFFLINE Status	Internet Problem
Missing DB Records	Server Problem
Repeated Reboots	Power Problem

Section 14 – Preventive Maintenance

14.1 Purpose

Preventive maintenance maximizes system reliability and minimizes downtime.

Focus areas include:

Inspection

Cleaning

Verification

Testing

Documentation

14.2 Recommended Maintenance Schedule

Activity	Interval
Dashboard Review	Weekly
Outdoor Inspection	Monthly

Solar Cleaning	Monthly
Sensor Inspection	Quarterly
Battery Inspection	Semiannual
Full Functional Test	Annual

14.3 Weekly Tasks

Verify:

Dashboard Updating

Outdoor Uptime Increasing

Weather Underground ONLINE

14.4 Monthly Tasks

Inspect:

Mast

Enclosure

Solar Panel

Mounting Hardware

Clean the solar panel as required.

14.5 Quarterly Tasks

Inspect:

Wind Vane

Anemometer

Rain Gauge

Sensor Wiring

Verify proper operation of each component.

14.6 Semiannual Tasks

Inspect:

Battery

Charge Controller

Power Wiring

Verify all connections remain secure.

14.7 Annual Tasks

Perform a complete functional verification.

Verify:

Sensors

LoRa

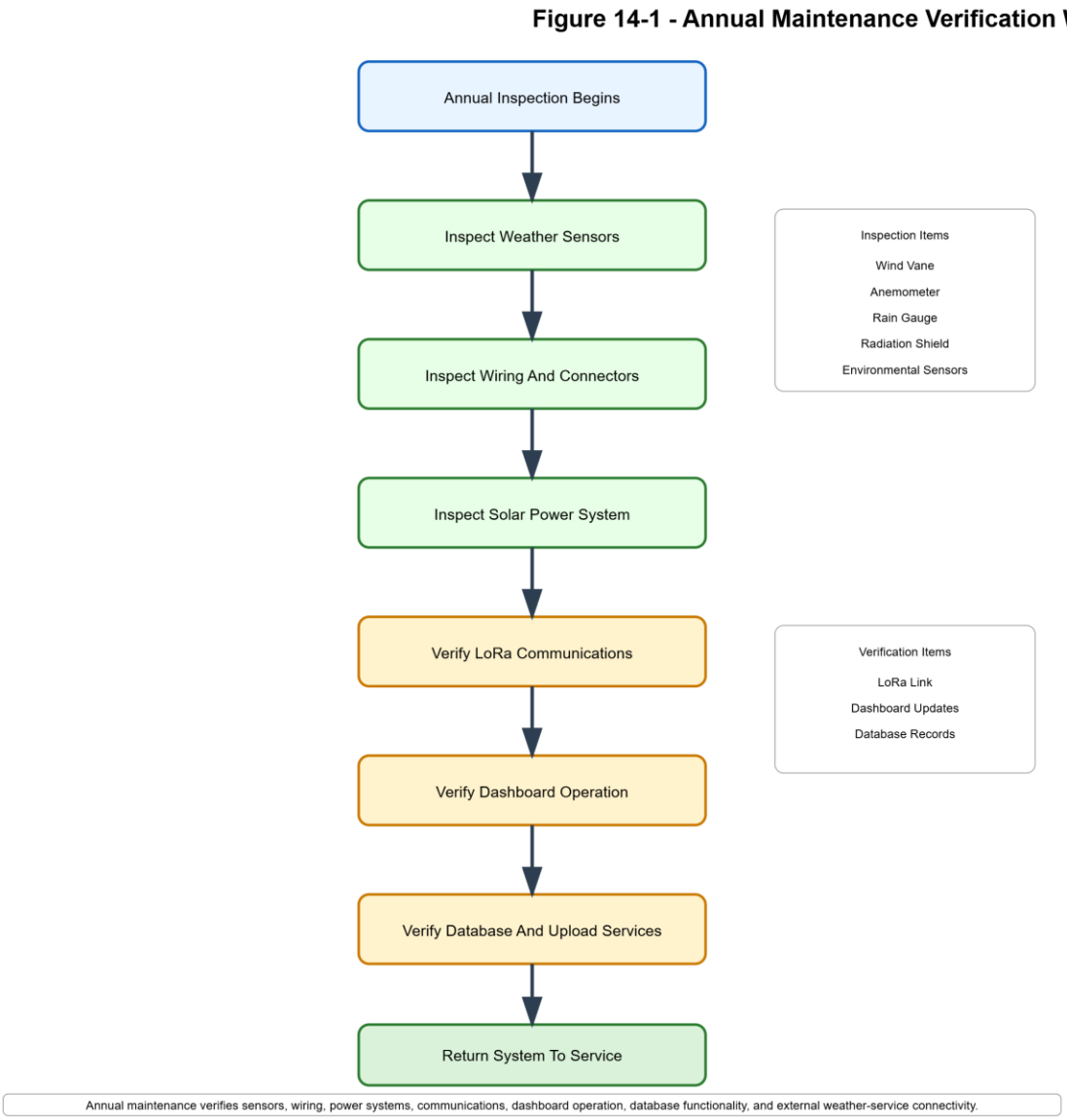
Dashboard

Database

Weather Underground

all remain operational.

Figure 14-1. Annual Maintenance Verification Flow



Annual maintenance verification sequence used to validate operation of all major system components.

14.8 Post-Storm Inspection

After severe weather:

Inspect:

Sensors

Pole

Enclosure

Solar Panel

Antenna

Cabling

and verify normal operation.

14.9 Service Log

A maintenance record should include:

Date

Technician

Findings

Corrective Actions

Firmware Version

Long-term records simplify support and system evolution.

Section 15 – Operations and Long-Term Support

15.1 Purpose

This section provides operational guidance that supplements the technical material presented throughout this manual. While previous sections focus on system architecture, firmware, communications, and hardware implementation, this section focuses on day-to-day operation, recovery procedures, preventive stewardship, and long-term maintenance.

The objective is to ensure that future maintenance personnel can successfully operate, troubleshoot, and preserve the Wireless Weather Station system even if the original designer is no longer available.

15.2 System Startup Procedure

Cold Start

After a complete power loss, the system should start in the following sequence:

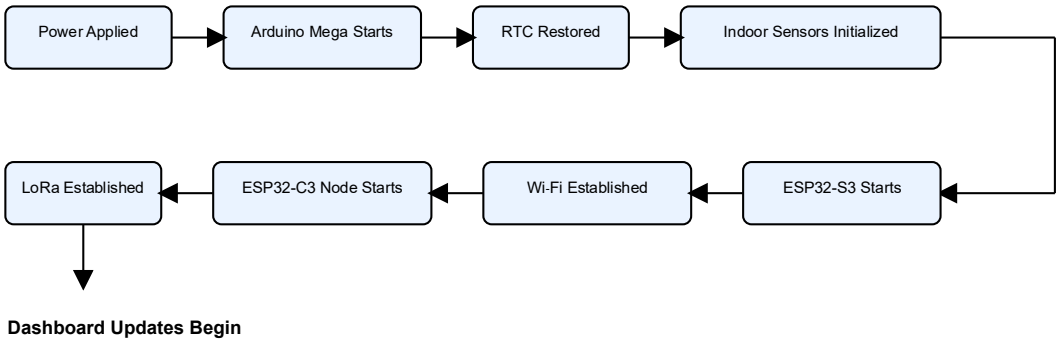


Figure 15-1. Cold Start Initialization Sequence

Normal startup should require no operator intervention.

Startup Verification Checklist

Verify:

- LCD active
- Date and time valid
- Ext OK displayed
- Outdoor uptime increasing
- Dashboard updating
- Weather Underground ONLINE
- Database records appearing

15.3 Normal Operations Checklist

The station is intended to operate continuously.

Arduino Mega

Verify:

- Display active
- Current weather displayed
- Button navigation functional

ESP32-C3 Outdoor Node

Verify:

- Outdoor uptime increasing
- Wind updates present
- Rain updates present

ESP32-S3 Gateway

Verify:

- Dashboard updating
- Database uploads active
- Weather Underground ONLINE

15.4 Recovery Procedures

Communications Failure Recovery

If outdoor communications stop:

1. Observe Ext NO condition.
2. Verify outdoor uptime.
3. Allow automatic recovery.
4. Observe STATUS requests.
5. Observe REBOOT attempts.
6. Verify uptime reset after recovery.

Dashboard Recovery

If dashboard updates stop:

1. Verify Arduino Mega operation.
2. Verify UART communications.
3. Verify ESP32-S3 operation.
4. Verify browser connectivity.
5. Restart ESP32-S3 if necessary.

Internet Recovery

If uploads fail:

1. Verify Wi-Fi connectivity.
2. Verify local Internet access.
3. Verify Weather Underground status.
4. Verify database connectivity.
5. Observe automatic recovery behavior.

15.5 Seasonal Inspection Recommendations

Spring

- Rain-gauge cleaning
- Wiring inspection
- Sensor inspection

Particular attention should be given to winter-weather damage.

Summer

- Solar-panel cleaning
- Radiation-shield inspection
- Temperature verification

Excessive contamination can affect temperature accuracy.

Fall

- Rain-gauge cleaning
- Leaf removal
- Sensor inspection

This period is particularly important for maintaining rainfall accuracy.

Winter

- Battery inspection
- Solar charging verification
- Enclosure-seal inspection

Cold-weather operation should be verified before severe weather arrives.

15.6 Documentation and Backup Strategy

The following records should be preserved throughout the life of the system:

Technical Documents

- Technical Manual
- User Manual
- Protocol Specifications
- Wiring Diagrams

Firmware Archives

- Arduino Mega firmware
- ESP32-S3 firmware
- ESP32-C3 firmware

All released versions should be archived.

Installation Records

- Installation photographs
- Wiring photographs
- Enclosure photographs

- Sensor alignment notes

These records significantly simplify future troubleshooting and upgrades.

Recommended Backup Schedule

Item	Recommended Frequency
Firmware	Before major changes
Technical Documentation	After each revision
Database	Weekly
Configuration Files	After modification

Historical weather data becomes increasingly valuable as the archive grows.

15.7 Spare Equipment Strategy

Recommended on-hand inventory:

Quantity	Item
1	Arduino Mega
1	ESP32-S3
1	ESP32-C3
1	LoRa Module
10	RJ11/RJ12 Connectors
2	Sensor Cables
1	LiFePO4 Battery
Assorted	Resistors and wiring components

The objective is to enable rapid recovery from common hardware failures.

15.8 Long-Term Support Philosophy

The weather station was designed around three primary goals:

- Reliability
- Serviceability
- Expandability

Future modifications should preserve these principles whenever practical.

New features should not compromise:

- Existing weather acquisition
- Existing communications
- Existing recovery mechanisms

The preferred approach is evolutionary improvement rather than architecture replacement.

Appendices A–H

Reference Material, Asset Inventory, Change History, and Future Planning

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Appendix A – GPIO and Signal Reference

A.1 Purpose

This appendix provides a consolidated reference for all controller GPIO assignments, communication interfaces, sensor connections, and signal paths used throughout the weather station.

The GPIO reference serves as the primary hardware reference for:

- Troubleshooting
- Firmware development
- Hardware expansion
- Maintenance
- System replication

A.2 Arduino Mega GPIO Assignments

User Interface

Pin	Function
A0	LCD button ladder input
D10	LCD backlight control

LoRa Communications (Serial2)

Pin	Function
D16	LoRa TX to RYLR890
D17	LoRa RX from RYLR890

ESP32-S3 Communications (Serial1)

Pin	Function
D18	UART1 TX to ESP32-S3
D19	UART1 RX from ESP32-S3

I^2C Bus

Pin	Function
D20	SDA
D21	SCL

A.3 ESP32-S3 GPIO Assignments

GPIO	Function
GPIO18	UART1 RX from Arduino Mega D18
GPIO17	UART1 TX to Arduino Mega D19

Serial Interface: UART1, 9600 baud.

A.4 ESP32-C3 GPIO Assignments

GPIO	Function
GPIO0	Wind vane ADC input
GPIO3	Rain gauge input
GPIO4	LoRa TX to RYLR890 RX
GPIO5	LoRa RX from RYLR890 TX
GPIO10	Anemometer input

A.5 Sensor Interface Summary

Indoor Sensors

Device	Interface
SHT31-D	I^2C
BMP280	I^2C
DS3231 RTC	I^2C

Outdoor Sensors

Device	Interface

Wind Vane	**ADC input**
Anemometer	**Interrupt input**
Rain Gauge	**Interrupt input**

A.6 Signal Path Summary

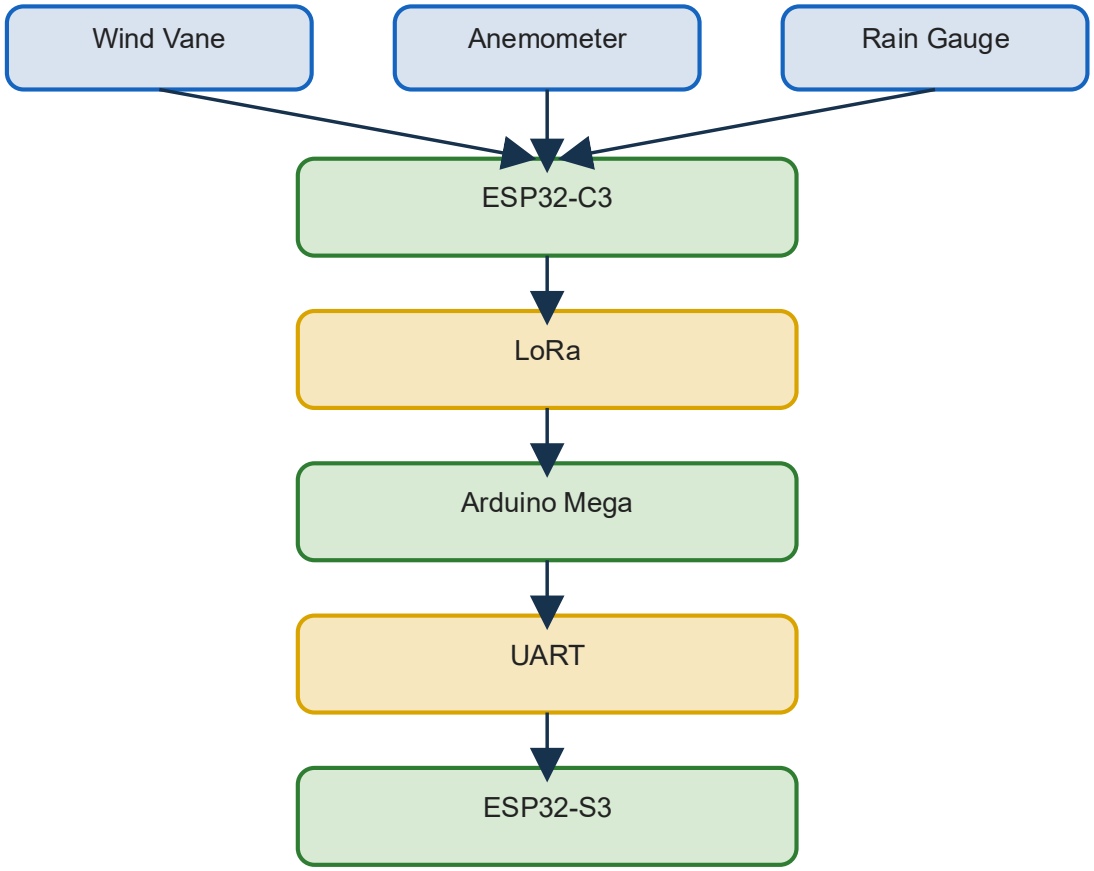


Figure A-1. Signal Path Summary

Consolidated signal flow from outdoor sensors through the ESP32-C3 outdoor controller, LoRa communications link, Arduino Mega supervisory controller, Serial1 UART interface, and ESP32-S3 gateway to dashboard, database, and Weather Underground services.

Appendix B – EEPROM Memory Map

B.1 Purpose

EEPROM storage preserves critical station information across power cycles.

Stored information includes:

- Environmental records
- Settings
- Reset information
- Calibration parameters

B.2 EEPROM Allocation

Address	Function
0	Min Temperature
4	Max Temperature
8	Min Humidity
12	Max Humidity
16	Min Pressure
20	Max Pressure
24	Altitude
28–48	Timestamp Records
500	Reset Flag
520–523	Last Reset Information

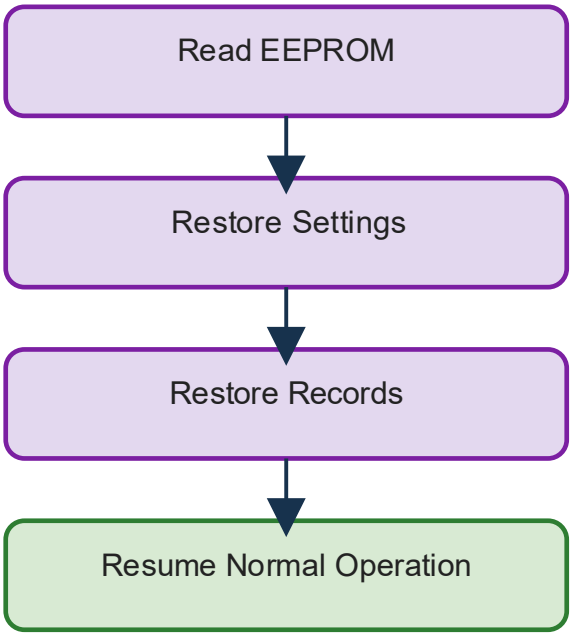
B.3 EEPROM Organization

EEPROM storage is organized into four logical regions:

- **Environmental Records** store minimum and maximum weather values.
- **Timestamps** store the associated date and time information.
- **Configuration** stores persistent operating parameters such as altitude.
- **Recovery Information** stores reset and restart data used for diagnostics.

B.4 Startup Recovery

Upon startup:



Appendix C – Communications Protocol Reference

C.1 Protocol Layers

Layer 1 LoRa

Layer 2 UART

Layer 3 HTTP

C.2 OUT Packet

Purpose

Primary outdoor telemetry packet.

Structure

<OUT>

WindSpeed,

WindGust,

Direction,

RainRate,

DailyRain,

RainTips,

Uptime

</OUT>

C.3 STATUS Protocol

Request

<STATUS?>

Response

<STATUS>

C.4 REBOOT Protocol

Request

<REBOOT>

Response

<REBOOTING>

C.5 UART Weather Packet

The Mega forwards:

Temperature

Humidity

Dew Point

Pressure

Wind Speed

Wind Direction

Rainfall

Outdoor Uptime

Date

Time

to the ESP32-S3.

C.6 Communications Timing

Activity	Interval
LoRa Telemetry	~5 sec
UART Update	~5 sec
Status Request	~30 sec
Reboot Trigger	~60 sec
Retry Count	3

Appendix D – Database Field Dictionary

D.1 Temperature Fields

Field	Description
temp	Corrected temperature
rawtemp	Raw sensor reading
adjfactor	Applied adjustment

D.2 Moisture Fields

Field	Description
humidity	Relative humidity
dewpt	Dew point
moist	Moisture classification
dewpot	Dew-potential classification

D.3 Pressure Fields

Field	Description
presshpa	Pressure in hPa
pressinhg	Pressure in inches Hg

D.4 Wind Fields

Field	Description
winddir	Direction
windspd	Wind speed
windgust	Peak gust
windlast	Previous direction value

D.5 Rain Fields

Field	Description
rainrate	Current rain intensity
rainday	Daily rainfall
raintips	Bucket-tip count

D.6 Health Fields

Field	Description
outuptime	ESP32-C3 uptime
nextupl	Upload timer

lastconn	Last connection
WUstat	Weather Underground status

Appendix E – System Parts List (BOM)

E.1 Indoor Controller

Qty	Component
1	Arduino Mega 2560
1	ESP32-S3
1	LCD Keypad Shield
1	SHT31-D
1	BMP280
1	DS3231 RTC
1	RYLR890 LoRa Module

E.2 Outdoor Node

Qty	Component
1	ESP32-C3
1	RYLR890 LoRa Module
1	Wind Vane
1	Anemometer
1	Tipping-Bucket Rain Gauge

E.3 Solar Power System

Qty	Component
1	Newpowa NPA5S-12J Solar Panel
1	SOLPERK 8A Charge Controller
1	12V LiFePO4 Battery

E.4 Spare Parts Inventory

Recommended stock:

ESP32-C3

ESP32-S3

LoRa Module

SHT31-D

BMP280

DS3231

RJ11/RJ12 Connectors

10K and 12K Resistors

Sensor Cable

Appendix F – Source File Inventory

F.1 Firmware Files

Arduino Mega

Min_Max_Temp_with_LoRa.ino

Purpose: Main Weather Controller

ESP32-S3

ESP32_S3_Inside_Station.ino

Purpose: Gateway / Dashboard

ESP32-C3

ESP32_C3_outside_station.ino

Purpose: Outdoor Telemetry Node

F.2 PHP Files

weatherdb.php

Purpose: Database Upload Handler

---

index.php

Purpose: Dashboard Interface

---

dewpoint.php

Purpose: Dew Point Calculations

F.3 Engineering References

New RJ11 and Wireless Connections.txt

Power Supply Info.txt

LoRa Output.txt

Appendix G – Engineering Notes and Design Rationale

G.1 Development Progression

The weather station evolved through several major phases:

Phase	Major Enhancement	Result
1	Single Controller	Initial weather monitoring platform with temp and humidity sensors
2	ESP32-S3 Gateway	Network connectivity and dashboard hosting
3	Web Dashboard	Remote monitoring
4	Outdoor Controller	Distributed data acquisition
5	LoRa Communications	Wireless outdoor telemetry
6	Database Storage	Historical weather archive
7	Solar Power System	Self-powered outdoor operation

G.2 Major Enhancements

Distributed Controller Architecture

Benefits:

- Reduced software complexity
- Better fault isolation
- Improved maintainability

LoRa Communications

Benefits:

- Reduced wiring

- Improved deployment flexibility
- Better sensor placement

Remote Reboot Capability

Benefits:

- Reduced service visits
- Improved uptime
- Automated recovery

Database Archiving

Benefits:

- Historical records
- Trend analysis
- Reporting capability

G.3 Why a Three-Controller Architecture Was Selected

The weather station could have been implemented using a single microcontroller.

That approach was ultimately rejected because it would have created a controller responsible for:

- Weather measurements
- User interface
- Database communications
- Dashboard hosting
- LoRa communications
- Internet communications

By distributing responsibilities across three processors, each subsystem remains relatively independent.

Arduino Mega

Responsible for:

- Weather Logic
- LCD Interface
- RTC
- EEPROM
- Recovery Management

ESP32-C3

Responsible for:

- Outdoor Data Acquisition
- LoRa Communications
- Rain Processing
- Wind Processing

ESP32-S3

Responsible for:

- Dashboard
- Wi-Fi
- Database Uploads
- Weather Underground Uploads

This separation significantly simplifies maintenance and troubleshooting.

G.4 Why LoRa Was Selected

Several wireless technologies were considered.

Wi-Fi

Advantages:

- High Throughput

- Easy Network Integration

Disadvantages:

- Higher Power Consumption
- Outdoor Coverage Challenges

Bluetooth

Advantages:

- Low Cost
- Simple Hardware

Disadvantages:

- Limited Range

LoRa

Advantages:

- Excellent Range
- Low Power Consumption
- Reliable Outdoor Operation
- Simple Deployment

LoRa was selected because it provided the best combination of range and reliability for the outdoor sensor node.

G.5 Why Solar Power Was Adopted

The original concept assumed permanent wired power.

During development it became apparent that outdoor sensor placement and wiring flexibility would improve significantly if the outdoor node became self-powered.

Benefits achieved:

- No power cable
- Flexible installation
- Reduced trenching

- Improved placement options

G.6 Why Outdoor Uptime Was Added

Outdoor uptime was not part of the original design.

Troubleshooting revealed that determining whether the outdoor node had rebooted was often difficult.

The uptime counter was introduced to provide visibility into:

- Unexpected Resets
- Power Problems
- Recovery Success
- Communications Health

It became one of the most valuable diagnostic metrics in the entire system.

G.7 Why Remote Reboot Was Added

Field failures occasionally required access to the outdoor node.

Because the node can be mounted outdoors and operate on solar power, physical access is less convenient than access to indoor equipment.

The remote reboot protocol was created so that:

- Communications Failures
- Software Lockups
- Temporary Faults

could often be recovered without a service visit.

G.8 Why a 12K Resistor Was Selected for the Wind Vane

Wind-Vane Signal Conditioning

The wind vane is connected to ESP32-C3 GPIO0 through a 12K pull-up resistor to the 3.3V rail.

The resistor forms a voltage divider with the vane's internal resistor network, allowing the ADC to distinguish individual wind-direction states.

A 12K pull-up resistor was selected through empirical testing. The value produced reliable separation between wind-direction ADC states while maintaining stable measurements. Testing showed that a 10K pull-up resistor did not provide sufficient separation between all vane positions.

G.9 Reliability Philosophy

The station was designed around a simple principle: Weather acquisition should continue whenever possible.

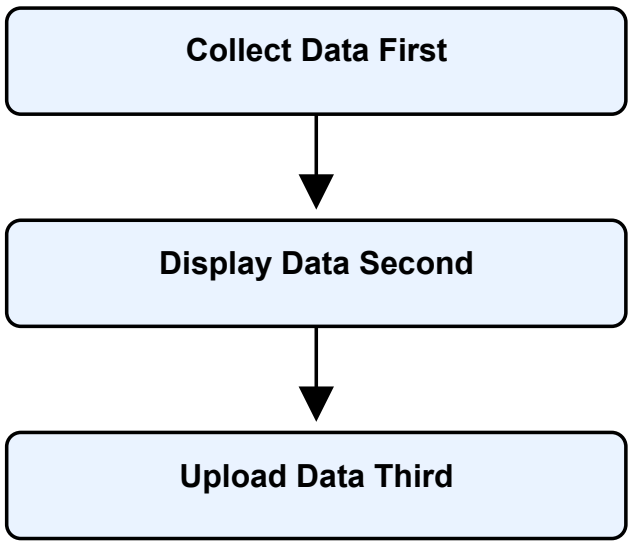


Figure G-1. Reliability Philosophy Hierarchy

Failures in:

- Internet Connectivity
- Database Services
- Dashboard Services

should not prevent weather collection.

This principle strongly influenced the final architecture.

G.10 Design Principles

The weather station follows the principles below.

Simplicity

Prefer understandable solutions over sophisticated solutions.

Recoverability

Subsystems should recover automatically whenever practical.

Visibility

Failures should be obvious.

Documentation

Critical behaviors should be documented.

Expandability

Future sensors should be easy to add.

G.11 Long-Term Support Objectives

The project was designed with an anticipated service life measured in years rather than months.

Long-term goals include:

- Minimal Maintenance
- Predictable Operation
- Simple Repair
- Incremental Enhancement

The technical manual exists partly to support these objectives.

Appendix H – Future Enhancements Roadmap

H.1 Power Monitoring

Future capabilities:

Battery Voltage

Charge Current

Solar Voltage

Battery Health

H.2 Environmental Expansion

Potential additions:

UV Index

Solar Radiation

Air Quality

H.3 Dashboard Improvements

Potential features:

Historical Graphs

Trend Analysis

Interactive Reporting

Mobile Optimization

H.4 Communications Enhancements

Potential features:

Packet Statistics

Reliability Metrics

Signal-Quality Tracking

Advanced Diagnostics

H.5 Remote Administration

Future possibilities:

Remote Configuration

Remote Calibration

Remote Firmware Updates

H.6 Version 2.0 Vision

Potential Version 2.0 capabilities:

Solar Telemetry

Battery Telemetry

Historical Charting

Advanced Forecasting

Remote Updates

Additional Sensors

Enhanced Analytics

while maintaining the proven architecture:

Arduino Mega

ESP32-S3 Gateway

ESP32-C3 Outdoor Node

LoRa Communications

Dashboard

Database
